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(54) **METHOD FOR SCREENING ALPHA BETA-TCR HETERODIMER MUTANTS**

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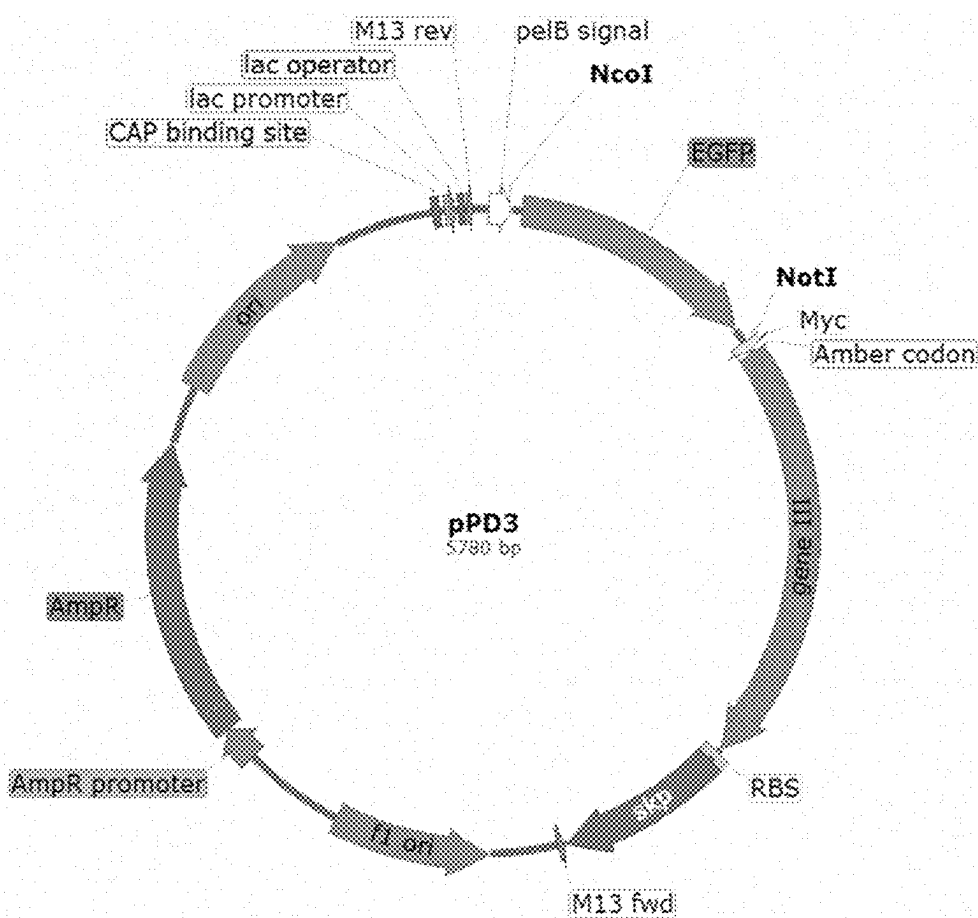
(57) **ABSTRACT**

A method for screening αβ-TCR mutants, the method comprising the construction of a phagemid. In the construction, a gene encoding an A1 polypeptide and a gene encoding B1 polypeptide are inserted into the 3' ends of genes encoding the constant regions of the α chain and the β chain of TCR respectively, and a ribosome binding site and a molecular chaperone gene are integrated into a phage vector, wherein the amino acid sequence of the A1 polypeptide is as shown in SEQ ID NO: 1 or is represented by the variant of the A1 polypeptide; the amino acid sequence of the B1 polypeptide is as shown in SEQ ID NO: 2 or is represented by the variant of the B1 polypeptide; and the variants at least retain the function of the polypeptides before mutation. The A1 peptide and the B1 peptide are used for promoting the correct pairing and folding of the αβ heterodimer TCR, so that the efficiency of displaying the correctly folded TCR on the surface of a phage is improved. The method can efficiently complete the screening of αβ heterodimer TCR mutants.

**Specification includes a Sequence Listing.**



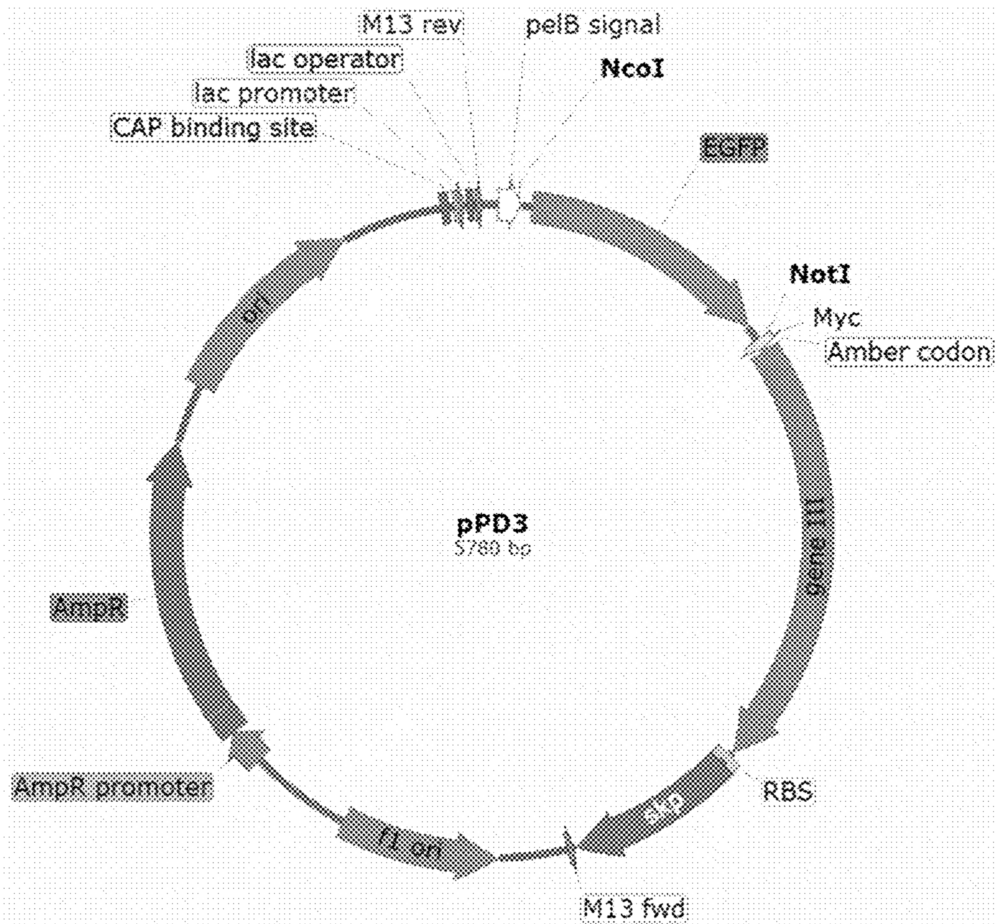

M13 rev
pelB signal
lac operator
NcoI
lac promoter
CAP binding site
EGFP
NotI
Myc
Amber codon
ori
pPD3
5780 bp
gene III
AmpR
RBS
AmpR promoter
f1 ori
M13 fwd

FIG. 1

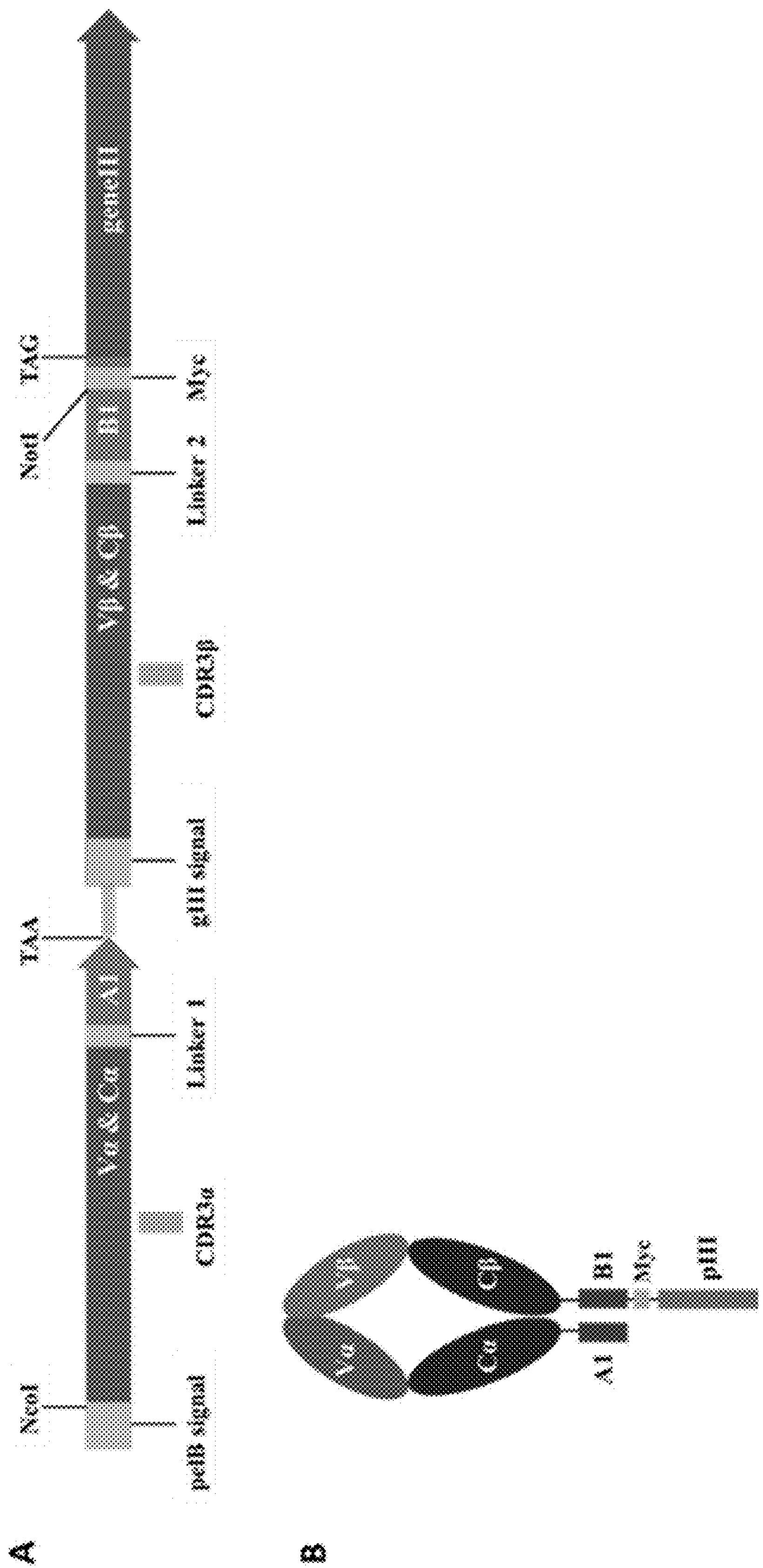

A
NcoI
pelB signal
CDR3α
Vα & Cα
Linker 1
A1
TAA
pIII signal
CDR3β
Vβ & Cβ
Linker 2
B1
NotI
Myc
TAG
geneIII
B
Vα
Vβ
Cα
Cβ
A1
B1
Myc
pIII

FIG. 2

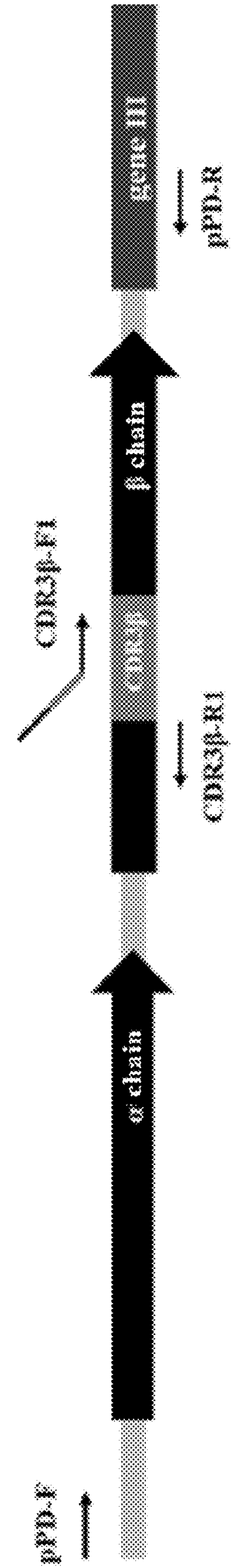



FIG. 3

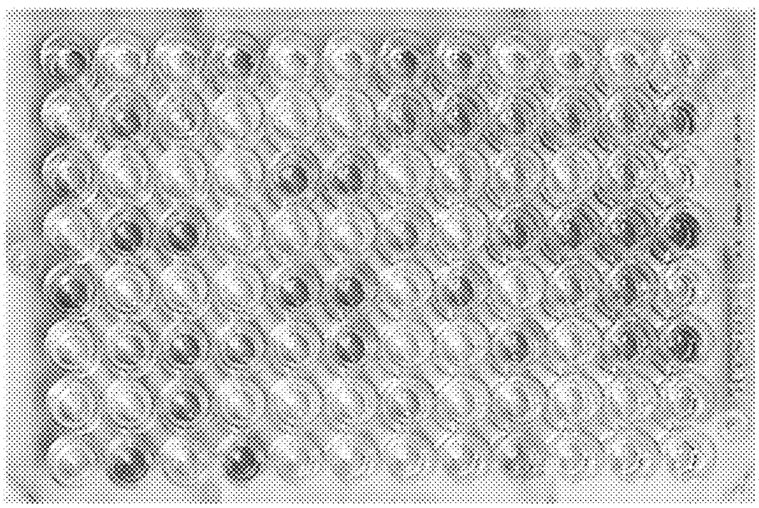


FIG. 4

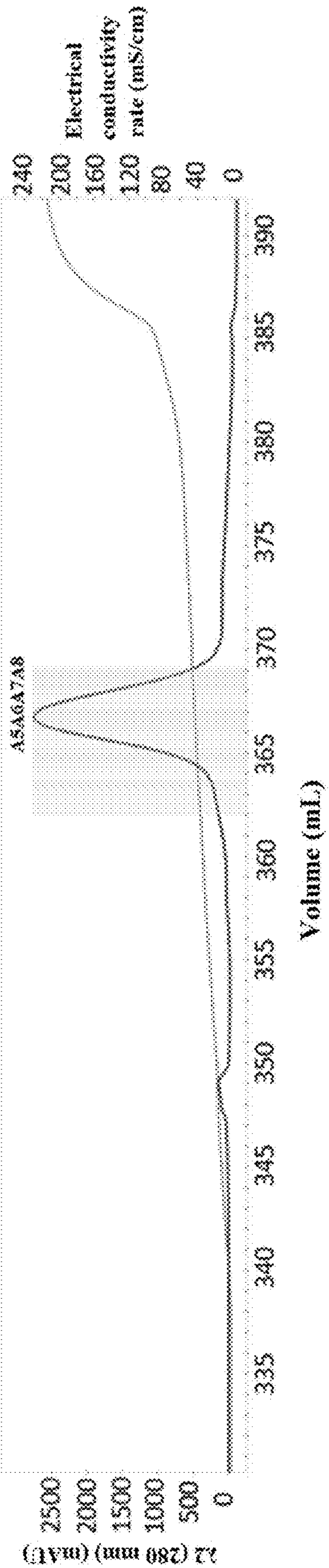

Electrical conductivity rate (mS/cm)
240
200
160
120
80
40
0
A5A6A7A8
335
340
345
350
355
360
365
370
375
380
385
390
Volume (mL)
λ2 (280 nm) (mAU)
2500
2000
1500
1000
500
0

FIG. 5A

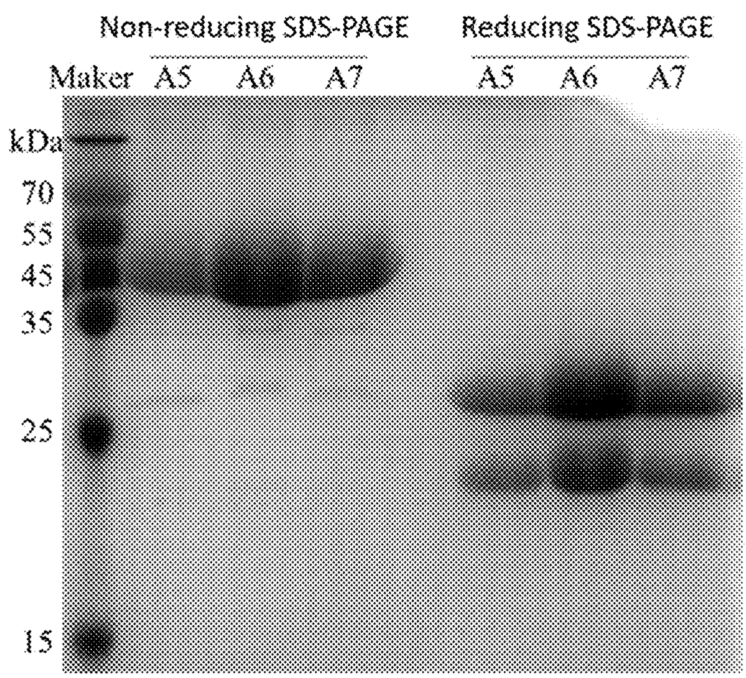

Non-reducing SDS-PAGE
Reducing SDS-PAGE
Maker
A5
A6
A7
A5
A6
A7
kDa
70
55
45
35
25
15

FIG. 5B

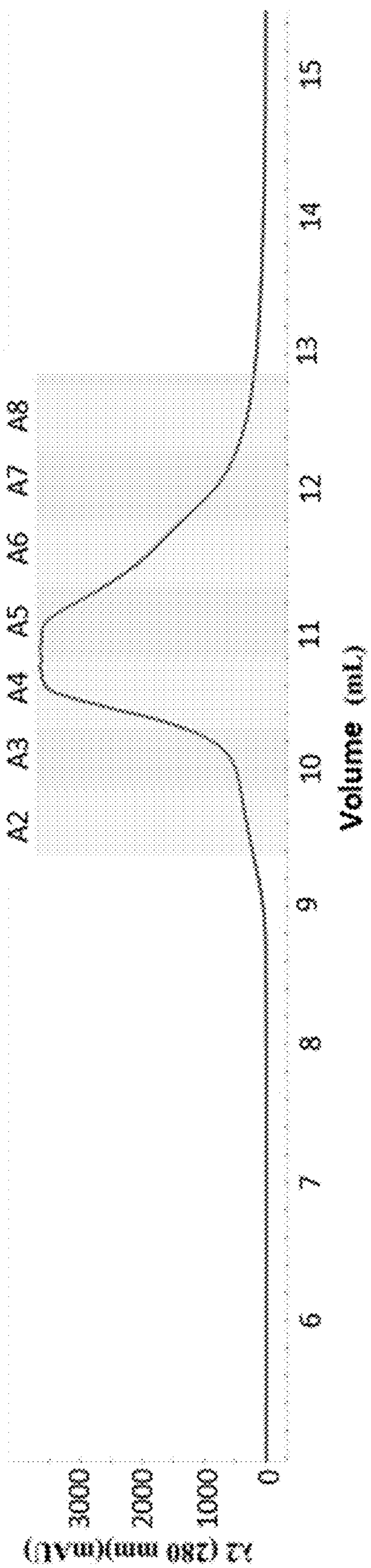

A2
A3
A4
A5
A6
A7
A8
3000
2000
1000
0
6
7
8
9
10
11
12
13
14
15
Volume (mL)

FIG. 6A

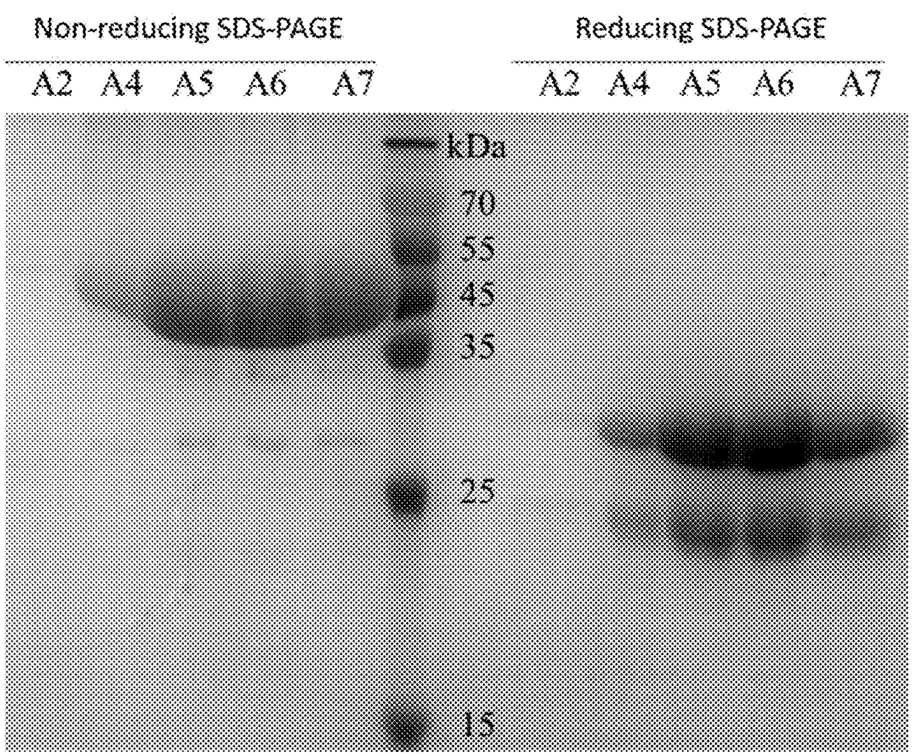

Non-reducing SDS-PAGE
Reducing SDS-PAGE
A2 A4 A5 A6 A7
A2 A4 A5 A6 A7
kDa
70
55
45
35
25
15

FIG. 6B

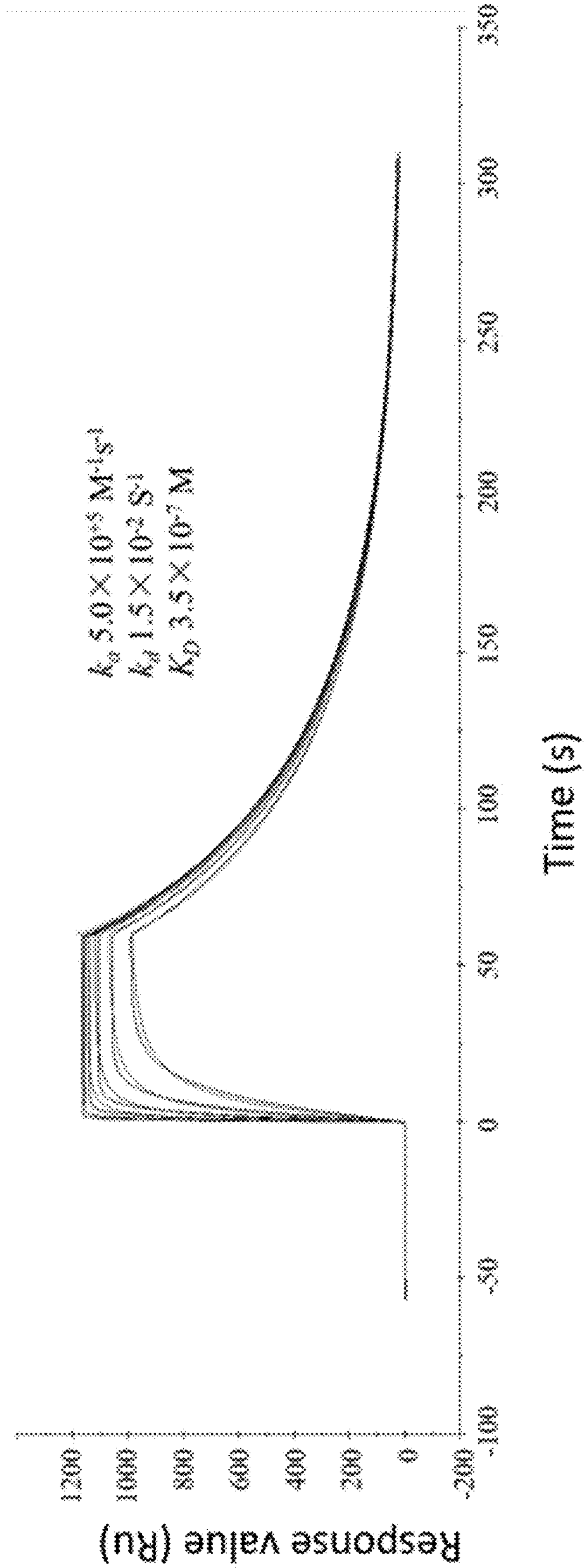

Response value (Ru)
1200
1000
800
600
400
200
0
-200
-100
-50
0
50
100
150
200
250
300
350
Time (s)
$k_a$ 5.0×10$^{-5}$ M$^{-1}$s$^{-1}$
$k_d$ 1.5×10$^{-2}$ S$^{-1}$
$K_D$ 3.5×10$^{-7}$ M

FIG. 7

# METHOD FOR SCREENING ALPHA BETA-TCR HETERODIMER MUTANTS

## REFERENCE TO SEQUENCE LISTING

**[0001]** The Replacement Sequence Listing is submitted concurrently with the specification as an ASCII formatted text file via EFS-Web, with a file name of "Replacement_Sequence_Listing_BSHING-23054-USPT.txt", a creation date of Oct. 30, 2023, and a size of 56,171 bytes. The Replacement Sequence Listing filed via EFS-Web is part of the specification and is incorporated in its entirety by reference herein.

## TECHNICAL FIELD

**[0002]** The present invention belongs to the field of biopharmaceutics, in particular relates to a method for screening αβ-TCR heterodimer mutants.

## BACKGROUND

**[0003]** In recent years, T cell receptor (TCR)-transduced T cell immunotherapy and TCR protein drugs have drawn increasing attention in the applications of antitumor therapy. Such therapies utilize TCR to specifically recognize the antigen peptide-major histocompatibility complex (pMHC) on the tumor surface. In the TCR-transduced T cell immunotherapy, TCR activates T cells to specifically kill tumor cells by specifically recognizing pMHC target on the tumor cell surfaces. It has been reported that TCRs with high affinity ($K_D$ is generally considered to be between 0.5-10 μM) may elicit more effective antitumor responses (*PNAS,* 2013, 110, 6973*; Immunology,* 2018, 155, 123). In terms of developing TCR protein drugs, the bivalent protein drug is currently the main form, which utilize TCR with ultra-high affinity to recognize the specific pMHC target on the tumor cell surface, and T cells are simultaneously activated by the anti-CD3 antibody at the same time, thereby achieving the specific killing of tumor cells (*Nat Med,* 2012, 18, 980*; Clin Cancer Res,* 2020, 26, 5869). Regardless of the technical pathway adopted, TCR molecules with high affinity are required.

**[0004]** Since T cells expressing TCRs with high-affinity in the thymus are eliminated during development (*Annu Rev Immunol,* 2003, 21, 139), the affinity of wild-type TCR to the pMHC is low in vivo, with a $K_D$ value typically only 1-500 μM (*Protein Eng Des Sel,* 2011, 24, 361). The improvement of the affinity of TCR to the pMHC is necessary for recognizing tumor-specific targets more effectively, thereby acting in the specific killing of tumor cells. Therefore, it is of significant importance to obtain the TCR with high affinity, high expression level, and high stability (*J Biol Chem,* 2005, 280, 1882*; Nat Med,* 2012, 18, 980*; Nat Commun,* 2019, 10, 4451*; Nat Commun,* 2020, 11, 2330; Cells 2020, 9, 1485) by in vitro engineering for the development of TCR therapies.

**[0005]** In vitro engineering of TCRs can be accomplished by the methods of yeast display (*Methods Mol Biol,* 2015, 1319, 95), phage display (*J Immunol Methods,* 1998, 221, 59*; Nat Biotechnol* 2005, 23, 349) or mammalian cell display (*J Biol Chem,* 2019, 294, 5790), and screening. The expression and display of TCR may utilize the two structures of single chain (Single-chain $V_\alpha$-linker-$V_\beta$, scTCR) and double chain (Disulfide-linked $V_\alpha C_\alpha/V_\beta C_\beta$, dsTCR) (*Biomol Eng,* 2007, 24, 361): scTCR, as a single-chain molecule, has the advantages of small molecular weight and high solubility. In the next place, scTCR avoids the structure of heterodimeric α- and β-chains, which is less prone to generate mismatched dimers and has higher efficiency in surface displays, such as phage/yeast display. However, due to the extreme instability of scTCR per se, some amino acid sites are needed to be mutated prior, to improve the stability of scTCR and the efficiency of correct folding, which increases the workload before display experiments and has a rather high failure rate. dsTCR, on the other hand, does not require stability optimization, and wild-type TCR may be subject to display experiments without concerning about the adverse effects, such as structural differences and immunogenicity, resulting from the transplantation of the mutation site from scTCR to αβ-TCR heterodimers (in vitro refolding/lentiviral packaging). Therefore, the use of dsTCR exhibits higher efficiency during affinity maturation. Establishing a dsTCR optimization platform with high performance is thus necessary. Nevertheless, in the absence of the transmembrane region and CD3 molecules, the folding and pairing efficiency of a and β chains of TCRs is extremely low, and it is difficult to obtain the αβ-TCR correctly folded and correctly paired during phage display. It is critical during phage display that how to improve the folding efficiency of TCRs and obtain αβ-TCRs correctly paired while using dsTCR for display experiments. Many studies on the solubility and folding efficiency of TCRs are available, such as the method to generate heterodimers by using the interchain disulfide bond between the constant regions of TCR (*Protein Eng,* 2003, 16, 707*; Nat Biotech,* 2005, 23, 349), which is used for the development of TCR therapy. In some studies, the coiled-coil sequence is introduced to promote the refolding of dsTCR in vitro (*Protein Sci,* 1999, 8, 2418*; J Immunol Methods,* 2000, 236, 147), which has not been applied in phage display. Co-expression of molecular chaperones to promote protein folding are also reported in some studies, such as promoting scFv/scTCR folding using skp (*J Immunol Methods,* 2005, 306, 51).

## SUMMARY

**[0006]** The technical problem to be solved by the present invention is to provide a method for screening an αβ-TCR mutant, in order to overcome the shortcomings of low efficiency of the methods used to screen effective TCR mutants or to engineer TCRs in the prior art. The screening of heterodimeric αβ-TCR mutants can be accomplished efficiently using the method of the present invention.

**[0007]** The present invention solves the above-mentioned technical problems mainly by the following technical solutions.

**[0008]** The first technical solution of the present invention is: a method for screening an αβ-TCR or a mutant thereof, comprising the construction of a phagemid, in the construction: the gene encoding A1 polypeptide and the gene encoding B1 polypeptide are inserted at the 3' ends of genes encoding constant regions of the α-chain and the β-chain of a TCR respectively, and a ribosomal binding site and a molecular chaperone gene are integrated into a phage vector;

**[0009]** the amino acid sequence of the A1 polypeptide is as shown in SEQ ID NO: 1 or a variant thereof, the amino acid sequence of the B1 polypeptide is as shown in SEQ ID NO: 2 or a variant thereof, the variant retains at least the function of the unmutated polypeptides.

[0010] The gene encoding the constant region of TCR α chain and the gene encoding the A1 polypeptide may be linked directly, or linked by a linker. Similarly, the gene encoding the constant region of TCR β chain and the gene encoding the B1 polypeptide may be linked directly, or linked by a linker. The two linkers used above may be conventional in the art, the sequences of which may be identical or different. In the present invention, the amino acid sequences of the two linkers are identical, preferably are as shown in SEQ ID NO: 3;

[0011] The constant region of TCR α chain and the constant region of TCR β chain in the present invention may be conventional in the art, such as the constant region of human TCR. In a specific embodiment of the present invention, the amino acid sequence of the constant region of the TCR α chain is as shown in SEQ ID NO: 5. The amino acid sequence of the constant region of the TCR β chain is as shown in SEQ ID NO: 4 or SEQ ID NO: 6, or the variant thereof, the variant is preferably obtained by substitution of position 75 with A and substitution of position 89 with D in the sequence of SEQ ID NO: 6.

[0012] The introduction of the ribosome binding site and the molecular chaperone gene are used to further promote the expression and folding of the αβ-TCR heterodimer, both are preferably located at the 3' end of the gene expressing a capsid protein. As is conventional in the art, the capsid protein is used to display the target protein, and the available capsid proteins are such as pIII protein, pVIII, etc. In the present invention, the pIII protein is preferred to display the target protein. Therefore, in a specific embodiment of the present invention, the ribosome binding site and the molecular chaperone gene are inserted at the 3' end of gene III.

[0013] The nucleotide sequence of the ribosome binding site may be conventional in the art, for example, its sequence is as shown in SEQ ID NO: 7.

[0014] The molecular chaperone gene may be selected as skp, FkpA, etc. skp is selected as the molecular chaperone in the present invention. The nucleotide sequence of the skp gene is preferably as shown in SEQ ID NO: 8.

[0015] The backbone of the phage vector in the present invention may be conventional in the art, such as pCANTAB5E.

[0016] Preferably, in the pCANTAB5E, a pelB signal peptide gene and/or a Myc tag gene are introduced to replace the sequence between the gIII signal and the Amber Stop Codon; more preferably, the nucleotide sequence between the pelB signal peptide gene and the Amber Stop Codon (including the both) is as shown in SEQ ID NO: 9.

[0017] myc is used in the present invention to facilitate the cleavage of trypsin to release the phage from the streptavidin magnetic beads. If triethylamine elution method is used, the myc tag may not be added.

[0018] In the present invention, the gene of the TCR α chain and the gene of the TCR β chain are preferably located in two different expression frames;

[0019] preferably, the phagemid comprises the expression element expressing following polypeptide (1) and polypeptide (2),

[0020] polypeptide (1): pelB signal peptide-variable region of TCR α chain-constant region of α chain -Linker 1-A1 peptide,

[0021] polypeptide (2): gIII signal peptide-variable region of TCR β chain-constant region of β chain-Linker 2-B1 peptide-pIII protein; wherein, the gene of B1 peptide and the gene expression pIII protein is preferably linked by Amber Stop Codon.

[0022] More preferably, the cysteine residue located behind the constant region of the TCR α chain and the constant region of the TCR β chain, and located in the N-terminal portion of the transmembrane region is subject to substitution or deletion.

[0023] pelB signal peptide used in the present invention is intended to assist in the localization of the TCR α chain to the periplasmic space, and those skilled in the art may use a similar signal peptide instead.

[0024] The second technical solution of the present invention is: a TCR obtained by screening with the method described in the first technical solution, the α chain of the TCR comprises:

[0025] (1) CDRs 1-3 or a variant thereof of the α chain variant region having the amino acid sequence as shown in SEQ ID NO: 10, wherein: the amino acid sequence of the CDR3 variant is preferably as shown in SEQ ID NO: 19;

[0026] (2) CDRs 1-3 or a variant thereof of the α chain variant region having the amino acid sequence as shown in SEQ ID NO: 14;

[0027] (3) CDRs 1-3 or a variant thereof of the α chain variant region having the amino acid sequence as shown in SEQ ID NO: 16; or

[0028] (4) CDRs 1-3 or a variant thereof of the α chain variant region having the amino acid sequence as shown in SEQ ID NO: 12, the amino acid sequence of the variant is preferably as shown in any one of SEQ ID NOs: 40-53.

[0029] The TCR as described above, wherein, the β chain preferably comprises:

[0030] (1) CDRs 1-3 or a variant thereof of a β chain variable region having the amino acid sequence as shown in SEQ ID NO: 11;

[0031] (2) CDRs 1-3 or a variant thereof of a β chain variable region having the amino acid sequence as shown in SEQ ID NO: 15; or

[0032] (3) CDRs 1-3 or a variant thereof of a β chain variable region having the amino acid sequence as shown in SEQ ID NO: 17; wherein:

[0033] For (1): the variant is obtained by position 6 and/or position 8 of CDR3 being substituted, position 6 is preferably substituted with Y or F, position 8 is preferably substituted with R, T or Q; preferably, the sequence of the variant is as shown in any one of SEQ ID NOs: 21-24.

[0034] For (2): the sequence of the variant is preferably as shown in any one of SEQ ID NOs: 26-30.

[0035] For (3): a mutation occurs in 1 site, 2 sites, 3 sites or 4 sites of positions 2-6 of CDR3; when the mutation occurs in two or more sites, the two or more sites are contiguous; position 3 is preferably mutated to A, position 4 is preferably mutated to Q, position 5 is preferably mutated to G, position 6 is preferably mutated to S, R or W; preferably, the sequence of the variant is as shown in any one of SEQ ID NOs: 32-39.

[0036] The third technical solution of the present invention is: a phage vector for screening αβ-TCR heterodimers or a mutant thereof, the phage vector comprises an expression element expressing following polypeptide (1) and polypeptide (2):

**[0037]** polypeptide (1): pelB signal peptide-variable region of the α chain-constant region of α chain -Linker 1-A1 peptide;

**[0038]** polypeptide (2): gIII signal peptide-variable region of the β chain-constant region of β chain-Linker 2-B1 peptide-pIII protein; wherein, the gene of B1 peptide and the gene expressing pIII protein are preferably linked by Amber Stop Codon.

**[0039]** As described above, wherein, the amino acid sequence of the constant region of the α chain is preferably as shown in SEQ ID NO: 5.

**[0040]** The amino acid sequence of the constant region of the β chain is preferably as shown in SEQ ID NO: 4 or SEQ ID NO: 6;

**[0041]** the amino acid sequence of the Linker 1 is preferably as shown in SEQ ID NO: 3.

**[0042]** The amino acid sequence of the Linker 2 is preferably as shown in SEQ ID NO: 3.

**[0043]** The amino acid sequence of A1 peptide is preferably as shown in SEQ ID NO: 1.

**[0044]** The amino acid sequence of B1 peptide is preferably as shown in SEQ ID NO: 2.

**[0045]** The backbone of the phage vector is preferably pCANTAB5E. During the vector construction, elements in pCANTAB5E may be deleted or adjusted in sequence as needed. During the application of the present invention, a pelB signal peptide gene and/or a Myc tag gene are preferably introduced to substitute the sequence between the gIII signal and the Amber Stop Codon. After the substitution, the nucleotide sequence between the pelB signal peptide gene and the Amber Stop Codon (including the both) is as shown in SEQ ID NO: 9.

**[0046]** A ribosomal binding site and a molecular chaperone gene are preferably integrated into the phage vector, such as, integrated at the 3' end of a gene expressing a capsid protein; the molecular chaperone gene is preferably a skp gene, more preferably the sequence as shown in SEQ ID NO: 8; the sequence of the ribosomal binding site is preferably as shown in SEQ ID NO: 7.

**[0047]** The amino acid sequence of the constant region of the α chain is preferably as shown in SEQ ID NO: 5;

**[0048]** the amino acid sequence of the constant region of the β chain is preferably as shown in SEQ ID NO: 4 or SEQ ID NO: 6, or a variant thereof, the variant of the sequence as shown in SEQ ID NO: 6 preferably contains mutation sites C75A and N89D.

**[0049]** The third technical solution of the present invention is: a use of the polypeptide as shown in SEQ ID NO: 1 or 2 in screening a TCR mutant.

**[0050]** During the use, the gene encoding the polypeptide is preferably placed at the 3' end of the gene of constant region of the α chain or the β chain of the TCR, respectively.

**[0051]** Preferably, the gene encoding the polypeptide is linked to the gene of the constant region of the α chain or the β chain of the TCR by a linker; the amino acid sequence of the linker is preferably as shown in SEQ ID NO: 3.

**[0052]** Based on the common knowledge in the art, each of the above preferred conditions may be combined in any way to obtain each preferred example of the present invention.

**[0053]** The reagents and raw materials used in the present invention are commercially available.

**[0054]** The positive progressive effect of the present invention is that

**[0055]** The present invention utilizes A1 peptide and B1 peptide to promote the correct pairing and folding of αβ heterodimeric TCR, which in turn enhances the display efficiency of correctly folded TCRs on the surface of phage. The present invention can efficiently accomplish the screening of the heterodimeric αβ-TCR mutants. The method can also be used to optimize the stability, water solubility, and affinity of TCRs.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0056]** FIG. **1** is the plasmid profile of the phage display vector.

**[0057]** FIG. **2**A is a design schematic diagram of TCR gene CDS, FIG. **2**B is a structure schematic diagram of TCR heterodimer.

**[0058]** FIG. **3** is a schematic diagram of the primer design method for library construction.

**[0059]** FIG. **4** is the results of ELISA assay after the second-round screening for TCR2.

**[0060]** FIG. **5**A and FIG. **5**B are the anion exchange chromatography and SDS-PAGE of TCR3-A0B1.

**[0061]** FIG. **6**A and FIG. **6**B are the gel filtration chromatography and SDS-PAGE.

**[0062]** FIG. **7** is the affinity test graph of TCR1-A0B4.

## DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

**[0063]** The present invention is further described below in combination with specific examples. It should be understood that these examples are only intended to illustrate the present invention and are not intended to limit the scope of the present invention. Furthermore, it should be understood that after reviewing the content taught by the present invention, a person skilled in the art may make various alterations or modifications to the present invention, and these equivalent forms also fall within the scope defined by the claims appended to the present application.

### Example 1: Design and Construction of the Vector for Phage Display

**[0064]** In order to display TCR in heterodimeric form on M13 phage, based on the phage display vector pCANTAB5E (purchased from Shanghai Huzhen Industrial Co., Ltd.), the main elements of the vector (ori, fl ori, ampicillin resistance gene and gene III, etc.) were retained, and the gene sequences of pelB signal peptide, Myc tag and GFP, etc. were introduced to replace the original DNA sequence between gIII signal and Amber Stop Codon (TAG), and a ribosome binding site and molecular chaperone skp gene were added following gene III to construct the phagemid vector pPD3 (see FIG. **1** for vector profile). Restriction sites for restriction endonucleases NcoI and NotI were added at the terminus of pelB and upstream gene III, respectively, for the ligation of TCR gene to the display vector.

**[0065]** Nucleotide sequence between pelB and Amber Stop Codon (underlined sequences are for pelB, GFP, Myc and Amber Stop Codon in order, and bolded sequences are restriction sites of NcoI and NotI):

(SEQ ID NO: 9)

atgaaatacctattgcctacggcagccgctggattgttattactcgcggc
ccagccgg**ccatgg**ctctaatagtgactaatagtgactaatagtgagtga
gcaagggcgaggagctgttcaccggggggtgcccatcctggtcgagctgg
acggcgacgtaaacggccacaagttcagcgtgtccggcgagggcgagggc
gatgccacctacggcaagctgaccctgaagttcatctgcaccaccggcaa
gctgcccgtgccctggcccaccctcgtgaccaccctgacctacggcgtgc
agtgcttcagccgctaccccgaccacatgaagcagcacgacttcttcaag
tccgccatgcccgaaggctacgtccaggagcgcaccatcttcttcaagga
cgacggcaactacaagacccgcgccgaggtgaagttcgagggcgacaccc
tggtgaaccgcatcgagctgaagggcatcgacttcaaggaggacggcaac
atcctggggcacaagctggagtacaactacaacagccacaacgtctatat
catggccgacaagcagaagaacggcatcaaggtgaacttcaagatccgcc
acaacatcgaggacggcagcgtgcagctcgccgaccactaccagcagaac
acccccatcggcgacggccccgtgctgctgcccgacaaccactacctgag
cacccagtccgccctgagcaaagaccccaacgagaagcgcgatcacatgg
tcctgctggagttcgtgaccgccgccgggatcactctcggcatggacgag
ctgtacaagtaatagtgactaatagtgactaatagtgat**gcggccgc**tgg
atccgaacagaaactgatcagcgaagaggatctggctgaatag

**[0066]** Nucleotide sequence of the ribosome binding site and skp gene (underlined sequence):

(SEQ ID NO: 7)

tttgtttaactttaagaaggagatatacatatgaaaaagtggttattagc
tgcaggtctcggtttagcactggcaacttctgctcaggcggctgacaaaa
ttgcaatcgtcaacatgggcagcctgttccagcaggtagcgcagaaaacc
ggtgtttctaacacgctggaaaatgagttcaaaggccgtgccagcgaact
gcagcgtatggaaaccgatctgcaggctaaaatgaaaaagctgcagtcca
tgaaagcgggcagcgatcgcactaagctggaaaaagacgtgatggctcag
cgccagacttttgctcagaaagcgcaggcttttgagcaggatcgcgcacg
tcgttccaacgaagaacgcggcaaactggttactcgtatccagactgctg
tgaaatccgttgccaacagccaggatatcgatctggttgttgatgcaaac
gccgttgcttacaacagcagcgatgtaaaagacatcactgccgacgtact
gaaacaggttaaataatttgtttaactttaagaaggaga

(SEQ ID NO: 8)

atgaaaaagtggttattagctgcaggtctcggtttagcactggcaacttc
tgctcaggcggctgacaaaattgcaatcgtcaacatgggcagcctgttcc
agcaggtagcgcagaaaaccggtgtttctaacacgctggaaaatgagttc
aaaggccgtgccagcgaactgcagcgtatggaaaccgatctgcaggctaa
aatgaaaaagctgcagtccatgaaagcgggcagcgatcgcactaagctgg
aaaaagacgtgatggctcagcgccagacttttgctcagaaagcgcaggct
tttgagcaggatcgcgcacgtcgttccaacgaagaacgcggcaaactggt
tactcgtatccagactgctgtgaaatccgttgccaacagccaggatatcg
atctggttgttgatgcaaacgccgttgcttacaacagcagcgatgtaaaa
gacatcactgccgacgtactgaaacaggttaaataa

## Example 2: Construction of Wild-Type TCR Phagemid

**[0067]** The intact heterodimeric TCR comprises an α chain variable region (Vα), an α chain constant region (Cα), a β chain variable region (Vβ) and a β chain constant region (Cβ). To express the α chain and β chain of TCR respectively, we designed and synthesize the TCR gene (the structure is shown in FIG. **2**A), referring to a PhD thesis from the Department of Pharmacy at Cardiff University (Liddy N. 2013, Molecular engineering of high affinity T-cell receptors for bispecific therapeutics). The engineered coiled-coil sequences A1 and B1 were introduced to promote the proximity of the two chains and thus folding to form TCR correctly. The α chain and β chain genes were designed to be in 2 expression frames: the A1 peptide gene was placed downstream Cα, and pelB signal peptide-Vα-Cα-Linker 1-A1 peptide were fused to express; the B1 peptide gene was placed downstream Cβ, and III signal peptide-Vβ-Cβ-Linker 2-B1 peptide-pIII protein were fused to express. The cysteine (C) that forms the disulfide bond behind the constant region (N-terminus of the transmembrane region) of both chains of the TCR was removed and no additional C was added to form the disulfide bond to reduce the misfolding caused by the dismatch of the disulfide bond; in addition, mutations of C75A and N89D were present in Cβ. The design schematic diagram of TCR gene is shown in FIG. **2**A.

**[0068]** Amino acid sequence of A1:

(SEQ ID NO: 1)

STTVAQLEEKVKTLRAQNYELKSRVQRLREQVAQLAS

**[0069]** Amino acid sequence of B1:

(SEQ ID NO: 2)

STSVDELQAEVDQLQDENYALKTKVAQLRKKVEKLSE

**[0070]** Amino acid sequence of Linker 1:

(SEQ ID NO: 3)

GGSGGGGG

**[0071]** Amino acid sequence of Linker 2:

(SEQ ID NO: 3)

GGSGGGGG

**[0072]** Amino acid sequence of α chain constant region:

SEQ ID NO: 5

(SEQ ID NO: 5)

IQNPDPAVYQLRDSKSSDKSVCLFTDFDSQTNVSQSKDSDVYITDKTVL
DMRSMDFKSNSAVAWSNKSDFACANAFNNSIIPEDTFFPSPESS

**[0073]** Amino acid sequence of β chain constant region (below are TRBC1 and TRBC2 in order):

(SEQ ID NO: 4)

EDLNKVFPPEVAVFEPSEAEISHTQKATLVCLATGFFPDHVELSWWVNG
KEVHSGVSTDPQPLKEQPALNDSRYCLSSRLRVSATFWQNPRNHFRCQV
QFYGLSENDEWTQDRAKPVTQIVSAEAWGRAD

(SEQ ID NO: 6)

EDLKNVFPPEVAVFEPSEAEISHTQKATLVCLATGFYPDHVELSWWVNG
KEVHSGVSTDPQPLKEQPALNDSRYCLSSRLRVSATFWQNPRNHFRCQV
QFYGLSENDEWTQDRAKPVTQIVSAEAWGRAD

**[0074]** The synthesized full-length TCR gene was double digested with NcoI and NotI, and ligated by T4 DNA ligase to the display vector which was already digested by the same restriction endonuclease. TG1 competent cells are transfected by chemical transformation. The correct phagemid was obtained by single clone sequencing analysis and used as PCR template for library construction.

**[0075]** Both of the two expression frames have low expression level under the regulation of the weak promoter: lac promoter and lac operator; due to the presence of the Amber Stop Codon TAG upstream gene III, the translation efficiency was reduced (translation as glutamine inefficiently in TG1 strain). The expression level of the fusion protein with β chain would be lower than that of the α chain. The low expression level of TCR gene may avoid the formation of homodimers, especially avoiding the formation of β-chain homodimer, and the β chain may have greater probability of monovalent display on the phage surface, which contributes to the formation of heterodimeric αβ-TCR. Fusion proteins are localized to the periplasmic space under the action of the pelB signal peptide and gIII signal peptide, respectively. The interaction of A1 peptide and B1 peptide promotes the binding of the α chain and the β chain, which in turn are folded to form heterodimeric αβ-TCR (shown in FIG. **2**B). The TCR participated in phage packaging in the bacterial periplasmic space to form the phages that display heterodimeric αβ-TCR on surface, which were released into the culture medium.

Example 3: Construction of Mutant Library

**[0076]** The designs of Example 1 and Example 2 are theoretically applicable to the modifications of CDR1, CDR2 and CDR3, but only CDR3 was validated. The selected 4 TCRs with 3 different targets were significantly enriched in amino acid sequences after CDR3 screening, and the mutant affinity was enhanced. Information about the targets, the germline origins and the sequences, etc. of the 4 TCRs are indicated below:

**[0077]** The targets and the germline origins of the 4 TCRs

-continued

TCR2-β chain variable region:
(SEQ ID NO: 13)
DTGVSQDPRHKITKRGQNVTFRCDPI**SEHNR**LYWYRQTLGQGPEFLTY**F**
**QNEAQ**LEKSRLLSDRFSAERPKGSFSTLEIQRTEQGDSAMYLC**ASSLYG**
**GGDTQY**FGPGTRLTVL

TCR3-α chain variable region:
(SEQ ID NO: 14)
GIQVEQSPPDLILQEGANSTLRCNFS**DSVNN**LQWFHQNPWGQLINLFY**I**
**PSGT**KQNGRLSATTVATERYSLLYISSSQTTDSGVYFC**AVEGDSNYQLI**
WGAGTKLIIKPD

TCR3-β chain variable region:
(SEQ ID NO: 15)
DTGVSQDPRHKITKRGQNVTFRCDPI**SEHNR**LYWYRQTLGQGPEFLTY**F**
**QNEAQ**LEKSRLLSDRFSAERPKGSFSTLEIQRTEQGDSAMYLC**ASRHLY**
**NEQF**FGPGTRLTVL

TCR4-α chain variable region:
(SEQ ID NO: 16)
AQKITQTQPGMFVQEKEAVTLDCTYD**TSDQSYG**LFWYKQPSSGEMIFLI
**YQGSYDEQN**ATEGRYSLNFQKARKSANLVISASQLGDSAMYFC**AMREPR**
**GSARQLT**FGSGTQLTVLPD

TCR4-β chain variable region:
(SEQ ID NO: 17)
DGGITQSPKYLFRKEGQNVTLSCEQN**LNHDA**MYWYRQDPGQGLRLIYYS
**QIVND**FQKGDIAEGYSVSREKKESFPLTVTSAQKNPTAFYLC**ASSQGIG**
**LAGGIYEQY**FGPGTRLTVT

**[0078]** The germline of the α chain constant regions of above 4 TCRs were originated from TRAC, and the germline of the β chain constant regions were originated from TRBC2 (see Example 2 for sequence details).

**[0079]** The library was constructed by introduction of the saturated mutation including 20 types of amino acid after PCR amplification with primers carrying simplex codons. The mutation primers with triplet codon NNK (5'→3', the

| TCR | HLA | Target | TRAV | TRBV |
|---|---|---|---|---|
| TCR1 | A*02:01 | WT1(RMFPNAPYL) (SEQ ID NO: 54) | TRAV1-2 | TRBV7-9 |
| TCR2 | A*02:01 | WT1(RMFPNAPYL) (SEQ ID NO: 54) | TRAV39 | TRBV7-9 |
| TCR3 | A*02:01 | AFP(FMNKFIYEI) (SEQ ID NO: 55) | TRAV22 | TRBV7-9 |
| TCR4 | A*02:01 | PRAME(VLDGLDVLL) (SEQ ID NO: 56) | TRAV14 | TRBV19 |

TCR1-α chain variable region:
(SEQ ID NO: 10)
GQNIDQPTEMTATEGAIVQINCTYQ**TSGFNG**LFWYQQHAGEAPTFLSY**N**
**VLDGL**EEKGRFSSFLSRSKGYSYLLLKELQMKDSASYLC**AVSNSGAGSY**
**QLT**FGKGTKLSVIPN

TCR1-β chain variable region:
(SEQ ID NO: 11)
DTGVSQDPRHKITKRGQNVTFRCDPI**SEHNR**LYWYRQTLGQGPEFLTY**F**
**QNE**AQLEKSRLLSDRFSAERPKGSFSTLEIQRTEQGDSAMYLC**ASSLGL**
**GEETQY**FGPGTRLLVL

TCR2-α chain variable region:
(SEQ ID NO: 12)
ELKVEQNPLFLSMQEGKNYTIYCNYS**TTSDR**LYWYRQDPGKSLESLFV**L**
**LSNGAV**KQEGRLMASLDTKARLSTLHITAAVHDLSATYFC**GGGADGLTF**
GKGTHLIIQPY

simple base codon N represents the bases A, T, C and G; and K represents the bases G and T) were used at each amino acid site of the optimized CDR3; as the electrotransformation efficiency of the TG1 transformation competent cells was approximately $10^8$-$10^9$, 4 NNKs were introduced into each mutation primer. The mutation of CDR3α was designed on the reverse primer, therefore it was MNN (5'→3', M represents the bases A and C), and the mutation of CDR30 was designed on the forward primer, therefore it was NNK (5'→3'). FIG. **3** is the design schematic diagram of the primer, wherein pPD-F and pPD-R are the universal primers designed for the vector, CDR3β-R1 is the reverse primer without mutation; CDR3β is the forward primer carrying the mutation, the 3' end of this primer may complementarily bind to the template sequence, the middle (gray bolded portion) is the NNK mutation region, and the 5' end is reversely complementary to CDR3β-R1 for overlap extension PCR amplification of the full-length TCR gene.

**[0080]** For TCR1, 5 mutation primers were designed for CDR3α and CDR30 according to the length of CDR. The primer names and sequences are listed in Table 1 (the lowercase letters in the primer sequences are bases that may match the template, and the bolded uppercase part is the saturated mutation region of this primer). The forward primer CDR3α-F1 was paired with the reverse primers CDR3α-R1 and CDR3α-R2, respectively, while the forward primer CDR3α-F2 was paired with the reverse primers CDR3α-R3, CDR3α-R4 and CDR3α-R5, respectively; CDR3β was used in the same way.

TABLE 1

Primers for TCR1 library construction

| primer | SEQ ID NO: | sequence |
|---|---|---|
| pPD-F | 57 | acacaggaaacagctatgac |
| pPD-R | 58 | acactgagtttcgtcaccag |
| CDR3α-F1 | 59 | ggttcctaccagctgaccttcg |
| CDR3α-R1 | 60 | aggtcagctggtaggaaccagcacc**MNNMNNMNNMNNMNN**gcacaggtagg acgcag |
| CDR3α-R2 | 61 | aggtcagctggtaggaacc**MNNMNNMNNMNNMNN**taccgcgcacaggtagg ac |
| CDR3α-F2 | 62 | ttcggtaaaggtaccaaactg |
| CDR3α-R3 | 63 | agtttggtacctttaccgaaggtcagctggta**MNNMNNMNNMNN**ggagttagatac cgc |
| CDR3α-R4 | 64 | agtttggtacctttaccgaaggtcag**MNNMNNMNNMNN**agcaccggagttagatac c |
| CDR3α-R5 | 65 | agtttggtacctttaccgaa**MNNMNNMNNMNN**ggaaccagcaccggag |
| CDR3β-R1 | 66 | acacaggtacatcgcagaatcg |
| CDR3β-F1 | 67 | ttctgcgatgtacctgtgt**NNKNNKNNKNNK**ggtctgggtgaagaaaccc |
| CDR3β-F2 | 68 | ttctgcgatgtacctgtgtgcatct**NNKNNKNNKNNK**ggtgaagaaacccagtac |
| CDR3β-R2 | 69 | cagggaagatgcacacaggtac |
| CDR3β-F3 | 70 | acctgtgtgcatcttccctg**NNKNNKNNKNNK**gaaacccagtactttggtc |
| CDR3β-F4 | 71 | acctgtgtgcatcttccctgggtctg**NNKNNKNNKNNK**cagtactttggtccgg |
| CDR3β-F5 | 72 | acctgtgtgcatcttccctgggtctgggtgaa**NNKNNKNNKNNK**tttggtccgggtac C |

**[0081]** The library was constructed using amplification methods of error prone PCR and overlap extension PCR successively to obtain the TCR gene carrying random mutations. Taking the pair of primers in FIG. **3** as an example, in the first round of PCR amplification 2 groups were divided: one group of PCR using primers pPD-F and CDR3β-R1 to amplify the fragment without mutation (about 1260 bp), and the other group using CDR3β-F1 and pPD-R to amplify the fragment carrying the mutation (about 810 bp); PCR amplification was performed using KOD FX (TOYOBO, KFX-101) in a system of 25 μL:

**[0082]** The first round of PCR amplification system

| Component | Volume (μL) |
|---|---|
| 2 × PCR Buffer for KOD FX | 12.5 |
| dNTPs (2.0 mM each) | 3 |
| primer F (10 μM) | 1 |

-continued

| Component | Volume (μL) |
|---|---|
| primer R (10 μM) | 1 |
| Template (100 pg/μL) | 1 |

-continued

| Component | Volume (μL) |
|---|---|
| KOD FX DNA Polymerase(1.0 U/μL) | 0.5 |
| $ddH_2O$ | Up to 25 |

**[0083]** PCR reaction procedure: 95° C. for 3 min, 95° C. for 10 sec, 58° C. for 30 sec, 68° C. for 90 sec, 68° C. for 5 min, and 4° C. for holding; 1.2% agarose gel electrophoresis was performed after PCR amplification, the target band was cut out, and agarose gel recovery kit (QIAGEN, 28706) was used to recover the fragments.

**[0084]** The second round of PCR was performed using vector universal primers pPD-F and pPD-R, and the two small fragments purified from the first round of PCR products were used as templates to amplify the TCR gene carrying the random mutation at the mutation site of this primer (about 2070 bp); PCR amplification system of 50 μL is used:

**[0085]** The second round of PCR amplification system

| Component | Volume (μL) |
|---|---|
| 2 × PCR Buffer for KOD FX | 25 |
| dNTPs (2.0 mM each) | 6 |
| primer F (10 μM) | 1.5 |
| primer R (10 μM) | 1.5 |
| Template (100 pg/μL) | 3 |
| KOD FX DNA Polymerase (1.0 U/μL) | 1 |
| $ddH_2O$ | Up to 25 |

**[0086]** PCR reaction procedure: 95° C. for 3 min, 95° C. for 10 sec, 58° C. for 30 sec, 68° C. for 120 sec, 68° C. for 5 min, and 4° C. for holding; 1% agarose gel electrophoresis was performed after PCR amplification, the target band was cut out, and agarose gel recovery kit (QIAGEN, 28706) was used to recover the fragments. The same method was used for PCR amplification with other mutation primers.

**[0087]** PCR amplified TCR gene and phage display vector were subject to double digestion respectively with NcoI-HF (NEB, R3193S) and NotI-HF (NEB, R3189L); digestion system of 50 μL is used:

**[0088]** System of double digestion

| Component | Volume (μL) |
|---|---|
| 10 × CutSmart 5 μL | 5 |
| NcoI-HF | 1 |
| NotI-HF | 1 |
| PCR fragment/display vector | 1.5 μg |
| $ddH_2O$ | Up to 50 |

**[0089]** A sufficient amount of digestion system was prepared according to the number of TCR fragments and vector requirements, and cultured at 37° C. for overnight; the target bands (TCR was about 1730 bp, display vector was about 4500 bp) were cut out after 1% agarose gel electrophoresis separation, and purified by agarose gel recovery kit along with concentration determination.

**[0090]** The vector and the fragment were ligated using T4 DNA ligase (NEB, M0202S) at a ratio of 1:3, and one ligation system was prepared for each TCR fragment; the system was cultured at 16° C. for overnight.

Ligation System

| Component | Volume (μL) |
|---|---|
| 10 × Buffer | 5 |
| Display vector | 200 ng |
| PCR Fragment | 200 ng |
| $T_4$ DNA Ligase | 1 |
| $ddH_2O$ | Up to 50 |

**[0091]** According to the different CDRs of ligation products, multiple reaction products of the same CDR were mixed, purified with the purification kit (TAKARA, 9761), eluted with sterilized deionized water, and electrotransformed into TG1 competent cells (Lucigen, 60502-2). The bacteria cells were collected after spreading the plate and culturing to complete the library construction.

Example 4: Screening of Mutant Library

**[0092]** The mutant library was subject to screening for 2-3 rounds using the phage display method.

1) Phage Packaging and Precipitation

**[0093]** Library bacteria were inoculated in 50 mL 2×YT medium containing 100 μg/mL ampicillin and 2% glucose, with $OD_{600}$ between 0.05-0.08 after inoculation, and cultured at 37° C., 220 rpm until $OD_{600}$=0.4-0.5. Helper phage M13K07 was added at a ratio of 20:1 (phage:bacteria), mixed well and kept at 37° C. for 30 min in stationary; the mixture was centrifuged at 3000 g for 10 min at room temperature, and then the supernatant was discarded. The pellets were resuspended with 50 mL 2×YT medium containing 100 μg/mL ampicillin and 50 μg/mL kanamycin, and cultured at 26° C., 200-220 rpm for overnight.

**[0094]** Bacterial cultures were centrifuged at 4° C./3000 g for 10 min. The supernatant was mixed with PEG/NaCl solution at a ratio of 4:1 and further centrifuged at 4° C./3000 g for 20 min after being kept on ice for 1 hour. The precipitates were collected and resuspended with 10 mL of PBS, centrifuged at 4° C./3000 g for 20 min. The supernatant was collected and added with 2.5 mL of PEG/NaCl solution, mixed well and kept on ice for 20-30 min; the mixture was centrifuged at 4° C./3000 g for 30 min, and the precipitates were resuspended with 1 mL of PBS to obtain the phage solution.

2) Phage Panning

**[0095]** The milk solution (500 μL, 3% w/v) was added to 500 μL of phage solution described above for 1 hour blocking. Then, an appropriate amount of biotinylation-labeled pMHC solution was added, and the mixture was rotationally incubated for 1 hour at room temperature to form the phage-pMHC complex. After the addition of 50 μL streptavidin magnetic beads and 30 minutes rotational incubation at room temperature, the magnetic bead-phage-pMHC complex were further achieved. The magnetic beads were adsorbed by a magnet, and washed for 3 rounds with 1.5% Milk-PBS, 0.1% PBST, and PBS respectively. Then 0.1 mg trypsin (Sigma, T1426) (final concentration 1 mg/mL) was added and rotationally incubated for 30 min at room temperature. Via magnet adsorption, the supernatant was collected and used to infect 10 mL of pre-cultured TG1 strain (OD600=0.4-0.5). The bacteria strain was spread on the plate containing 100 μg/mL ampicillin and 2% glucose and cultured upside down at 32° C. for overnight. All the bacteria pellets on the plate were collected and stored. An appropriate amount of single clones on the plate for gradient dilution counting were selected to perform sequencing analysis.

**[0096]** The above process was repeated for 2-3 rounds of screening according to the experimental requirements, and the 2nd round or the 3rd round of screening reduced the pMHC concentration appropriately to obtain higher affinity sequences according to the degree of library enrichment.

**[0097]** The optimization of the CDR3 of 4 different TCRs of the present invention all achieved the results of significant enrichment of amino acid sequences. CDR3α and CDR3β were optimized in TCR1, wherein CDR3α sequence was not significantly enriched, and still the mutation sites were basically present in the whole region of CDR after screening (see Table 2), which may be resulted from the amino acid mutation in this CDR contributes less to its affinity enhancement to pMHC. It can be seen from the sequencing results of the 1st round screening of CDR3β that G97 and G99 were enriched (in clones CDR3β-11, 12 and 13, introduced by primer CDR3β-F3, and 4 amino acids of G97-E100 were mutated); in the sequencing results of the 2nd round screening, it presents more obvious enrichment, with all L98 mutated to Y and E100 mutated to T/Q/A, while all other sites were wild-type (see Table 3).

TABLE 2

Sequence analysis for library construction of TCR1 CDR3α

| Clone Type | Clone Number | CDR3α $_{(A89-T101)}$ | SEQ ID NO: |
|---|---|---|---|
| Wild type | TCR1-3α-wt | AVSNSGAGSYQLT | 18 |
| *E.coli* library clones | TCR1-3α-1 | **PDRPE**GAGSYQLT | 73 |
| | TCR1-3α-2 | AV**PLGTL**GSYQLT | 74 |
| | TCR1-3α-3 | AV**PKVGT**GSYQLT | 75 |
| | TCR1-3α-4 | AVSNS**RGKD**YQLT | 76 |
| | TCR1-3α-5 | AVSNSGA**EWAA**LT | 77 |
| | TCR1-3α-6 | AVSNSGAGS**SSRS** | 78 |
| | TCR1-3α-7 | AVSNSGAGS**NVND** | 79 |
| Clones produced by the 1st round screening | TCR1-3α-8 | **S*GTK**GAGSYQLT | 80 |
| | TCR1-3α-9 | AV**CFAMP**GSYQLT | 81 |
| | TCR1-3α-10 | AV***IYHT**GSYQLT | 82 |
| | TCR1-3α-11 | AV**DDLPN**GSYQLT | 83 |
| | TCR1-3α-12 | AVSNS**FPAT**YQLT | 84 |
| | TCR1-3α-13 | AVSNS**LNLP**YQLT | 85 |
| | TCR1-3α-14 | AVSNS**LAPP**YQLT | 86 |
| | TCR1-3α-15 | AVSNSG**R****YQLT | 87 |
| | TCR1-3α-16 | AVSNSG**LPR**YQLT | 88 |
| | TCR1-3α-17 | AVSNSGAG**ART**LT | 89 |
| | TCR1-3α-18 | AVSNSGA**LDH**SLT | 90 |
| | TCR1-3α-19 | AVSNSGA**QVAR**LT | 91 |
| | TCR1-3α-20 | AVSNSGGGS**NGWA** | 92 |
| Clones produced by the 2nd round screening | TCR1-3α-21 | AV**RCTQP**GSYQLT | 93 |
| | TCR1-3α-22 | AVSNS**FVEL**YQLT | 94 |
| | TCR1-3α-23 | AVSNS**QRVL**YQLT | 95 |
| | TCR1-3α-24 | AVSNSGA**S*CV**LT | 96 |
| | TCR1-3α-25 | AVSNSGA**YVFP**LT | 97 |
| | TCR1-3α-26 | AVSNSGAGS**RRWE** | 98 |

Note:
the nucleotide sequence corresponding to * is Amber Stop Codon TAG

TABLE 3

Sequence analysis for library construction of TCR1 CDR3β

| Clone Type | Clone Number | CDR3β $_{(A93-Y104)}$ | SEQ ID NO: |
|---|---|---|---|
| Wild type | TCR1-3β-wt | ASSLGLGEETQY | 20 |
| *E.coli* library clones | TCR1-3β-1 | **PQEP**GLGEETQY | 99 |
| | TCR1-3β-2 | **TDM**LGLGEETQY | 100 |
| | TCR1-3β-3 | **AP*T**GLGEETQY | 101 |
| | TCR1-3β-4 | AS**PRA**LGEETQY | 102 |
| | TCR1-3β-5 | ASSL**PTK**EETQY | 103 |
| | TCR1-3β-6 | ASSLGL**LVYL**QY | 104 |
| | TCR1-3β-7 | ASSLGL**Q*LA**QY | 105 |

TABLE 3-continued

Sequence analysis for library construction of TCR1 CDR3β

| Clone Type | Clone Number | CDR3β (A93-Y104) | SEQ ID NO: |
|---|---|---|---|
| Clones produced by the 1st round screening | TCR1-3β-8 | AS**GMVS**GEETQY | 106 |
| | TCR1-3β-9 | AS***TYA**GEETQY | 107 |
| | TCR1-3β-10 | ASSL**RELI**ETQY | 108 |
| | TCR1-3β-11 | ASSLG**MGR**ETQY | 109 |
| | TCR1-3β-12 | ASSLG**FGH**ETQY | 110 |
| | TCR1-3β-13 | ASSLG**YGS**ETQY | 111 |
| | TCR1-3β-14 | ASSLGL**GEET**QY | 20 |
| | TCR1-3β-15 | ASSLGL**PPGA**QY | 112 |
| | TCR1-3β-16 | ASSLGL**WRRR**QY | 113 |
| | TCR1-3β-17 | ASSLGL**GREI**QY | 114 |
| | TCR1-3β-18 | ASSLGLG**YDY**QY | 115 |
| | TCR1-3β-19 | ASSLGLG**CLV**QY | 116 |
| Clones produced by the 2nd round screening | TCR1-3β-20 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-21 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-22 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-23 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-24 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-25 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-26 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-27 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-28 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-29 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-30 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-31 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-32 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-33 | ASSLG**Y**G**T**ETQY | 23 |
| | TCR1-3β-34 | ASSLG**Y**G**Q**ETQY | 24 |
| | TCR1-3β-35 | ASSLG**Y**G**A**ETQY | 117 |

Note:
the nucleotide sequence corresponding to * is Amber Stop Codon TAG

**[0098]** The sequence enrichment of CDR3α of TCR2 was obvious after two rounds of phage display library screening: concentrated in the first four amino acid sites G90 for G/A, G91 mainly for G/A/V, G92 remained wild type, A93 mainly mutated for V/E, etc. of CDR3α. The sequencing analysis results are shown in Table 4.

TABLE 4

Sequence analysis for library construction of TCR2 CDR3α

| Clone Type | Clone Number | CDR3α (G90-T97) | SEQ ID NO: |
|---|---|---|---|
| Wild type | TCR2-3α-wt | GGGADGLT | 118 |
| *E.coli* library clones | TCR2-3α-1 | **RMIV**DGLT | 119 |
| | TCR2-3α-2 | **KNGL**DGLT | 120 |
| | TCR2-3α-3 | **ALW**ADGLT | 121 |
| | TCR2-3α-4 | GGGA**AVDG** | 122 |
| | TCR2-3α-5 | GGGA**MPPG** | 123 |
| | TCR2-3α-6 | GGGA**HVKW** | 124 |
| | TCR2-3α-7 | GGGAD**IWS** | 125 |
| Clones produced by the 1st round screening | TCR2-3α-8 | **YG*L**DGLT | 126 |
| | TCR2-3α-9 | ***PAE**DGLT | 127 |
| | TCR2-3α-10 | **GVGT**DGLT | 128 |
| | TCR2-3α-11 | GG**AHPS**LT | 129 |
| | TCR2-3α-12 | GG**AAVW**LT | 130 |
| | TCR2-3α-13 | GG**YVKL**LT | 131 |
| | TCR2-3α-14 | GG***LRW**LT | 132 |
| | TCR2-3α-15 | GGGA**EIVA** | 133 |
| Clones produced by the 2nd round screening | TCR2-3α-16 | **QKAE**DGLT | 134 |
| | TCR2-3α-17 | **ALGI**DGLT | 135 |
| | TCR2-3α-18 | **AA**G**V**DGLT | 46 |
| | TCR2-3α-19 | **AA**G**V**DGLT | 46 |
| | TCR2-3α-20 | **AG**G**V**DGLT | 45 |
| | TCR2-3α-21 | **G**V**GT**DGLT | 128 |

TABLE 4-continued

Sequence analysis for library construction of TCR2 CDR3α

| Clone Type | Clone Number | CDR3α (G90-T97) | SEQ ID NO: |
|---|---|---|---|
| | TCR2-3α-22 | GV**G**S**DGLT | 136 |
| | TCR2-3α-23 | GI**G**D**DGLT | 40 |
| | TCR2-3α-24 | GV**G**E**DGLT | 137 |
| | TCR2-3α-25 | GV**G**E**DGLT | 137 |
| | TCR2-3α-26 | GG**DIT***LT | 138 |
| | TCR2-3α-27 | GGGA**GWM*** | 139 |
| | TCR2-3α-28 | GGGAD**HFC** | 140 |

Note:
the nucleotide sequence corresponding to * is Amber Stop Codon TAG

[0099] CDR3β of TCR3 was significantly enriched, and the sequencing results of the 2nd round screening show that the mutation sites were mainly at 5 amino acid sites from Q95 to N99, and the results are shown in Table 5.

TABLE 5

Sequence analysis for library construction of TCR3 CDR3β Stop Codon TAG

| Clone Type | Clone Number | CDR3β (A93-F102) | SEQ ID NO: |
|---|---|---|---|
| Wild type | TCR3-3β-wt | ASRHLYNEQF | 25 |
| *E. coli* library clones | TCR3-3β-1 | **TLQF**LYNEQF | 141 |
| | TCR3-3β-2 | **EGKW**LYNEQF | 142 |
| | TCR3-3β-3 | **PHFL**LYNEQF | 143 |
| | TCR3-3β-4 | **RTVE**LYNEQF | 144 |
| | TCR3-3β-5 | AS**VOTT**NEQF | 145 |
| | TCR3-3β-6 | AS**PYD***NEQF | 146 |
| | TCR3-3β-7 | ASRH**SR*C**QF | 147 |
| | TCR3-3β-8 | ASRHLY**FDNG** | 148 |
| Clones produced by the 1st round screening | TCR3-3β-9 | **CSNT**LYNEQF | 149 |
| | TCR3-3β-10 | **R******P**LYNEQF | 150 |
| | TCR3-3β-11 | **ERRP**LYNEQF | 151 |
| | TCR3-3β-12 | AS**WNVS**NEQF | 152 |
| | TCR3-3β-13 | AS***LLC**NEQF | 153 |
| | TCR3-3β-14 | AS**YVVS**NEQF | 154 |
| | TCR3-3β-15 | AS**HPAR**NEQF | 155 |
| | TCR3-3β-16 | AS**ANAV**NEQF | 156 |
| | TCR3-3β-17 | AS**GRVR**NEQF | 157 |
| | TCR3-3β-18 | ASRHLY**SSGR** | 158 |
| | TCR3-3β-19 | ASRHLY**PLYA** | 159 |
| Clones produced by the 2nd round screening | TCR3-3β-20 | AS**QSW**YNEQF | 26 |
| | TCR3-3β-21 | AS**QRW**YNEQF | 27 |
| | TCR3-3β-22 | ASR**REF**NEQF | 28 |
| | TCR3-3β-23 | ASR**REF**NEQF | 28 |
| | TCR3-3β-24 | ASR**REF**NEQF | 28 |
| | TCR3-3β-25 | ASR**REF**NEQF | 28 |
| | TCR3-3β-26 | ASR**RSF**NEQF | 160 |
| | TCR3-3β-27 | ASR**LLF**NEQF | 30 |
| | TCR3-3β-28 | ASRH**EFD**EQF | 29 |
| | TCR3-3β-29 | ASRH**EFD**EQF | 29 |
| | TCR3-3β-30 | ASRH**EFD**EQF | 29 |
| | TCR3-3β-31 | ASRHLY**PGAL** | 161 |

Note:
the nucleotide sequence corresponding to * is Amber Stop Codon TAG

**[0100]** CDR3β of TCR4 was significantly enriched: it can be seen from the sequencing results of the 2nd round screening that S94 is S/A, Q95 is I/V/L, G96 is T, 197 is R, etc.; the results are shown in Table 6.

TABLE 6

Sequence analysis for library construction of TCR4 CDR3β

| Clone Type | Clone Number | CDR3β (A92-F107) | SEQ ID NO: |
|---|---|---|---|
| Wild type | TCR4-3β-wt | ASSQGIGLAGGIYEQY | 31 |
| *E.coli* library clones | TCR4-3β-1 | **NLDD**GIGLAGGIYEQY | 162 |
| | TCR4-3β-2 | **SCWI**GIGLAGGIYEQY | 163 |
| | TCR4-3β-3 | ***PFL**GIGLAGGIYEQY | 164 |
| | TCR4-3β-4 | AS**GHR**IGLAGGIYEQY | 165 |
| | TCR4-3β-5 | ASSQ**CGV**LAGGIYEQY | 166 |
| | TCR4-3β-6 | ASSQGIGL**PE**G**R**YEQY | 167 |
| | TCR4-3β-7 | ASSQGIGLAGGI**ERSL** | 168 |
| | TCR4-3β-8 | ASSQGIGLAGGI**RSKI** | 169 |
| Clones produced by the 1st round screening | TCR4-3β-9 | AS**ALTR**GLAGGIYEQY | 170 |
| | TCR4-3β-10 | AS**ALSY**GLAGGIYEQY | 171 |
| | TCR4-3β-11 | AS**AITD**GLAGGIYEQY | 172 |
| | TCR4-3β-12 | ASS**VTY**GLAGGIYEQY | 173 |
| | TCR4-3β-13 | ASS**ITS**GLAGGIYEQY | 174 |
| | TCR4-3β-14 | ASS**VTL**GLAGGIYEQY | 175 |
| | TCR4-3β-15 | ASS**LTY**GLAGGIYEQY | 176 |
| | TCR4-3β-16 | ASS**MTR**GLAGGIYEQY | 177 |
| | TCR4-3β-17 | ASS**VTM**GLAGGIYEQY | 178 |
| | TCR4-3β-18 | ASS**ITR**GLAGGIYEQY | 179 |
| | TCR4-3β-19 | ASS**QSR**GLAGGIYEQY | 180 |
| | TCR4-3β-20 | ASSQ**TSKC**AGGIYEQY | 181 |
| Clones produced by the 2nd round screening | TCR4-3β-21 | AS**AITR**GLAGGIYEQY | 182 |
| | TCR4-3β-22 | AS**AVTR**GLAGGIYEQY | 183 |
| | TCR4-3β-23 | AS**ALTS**GLAGGIYEQY | 184 |
| | TCR4-3β-24 | AS**ALTW**GLAGGIYEQY | 35 |
| | TCR4-3β-25 | ASS**LTI**GLAGGIYEQY | 32 |
| | TCR4-3β-26 | ASS**ITR**GLAGGIYEQY | 179 |
| | TCR4-3β-27 | ASS**ITR**GLAGGIYEQY | 179 |
| | TCR4-3β-28 | ASS**VTR**GLAGGIYEQY | 185 |
| | TCR4-3β-29 | ASS**LTR**GLAGGIYEQY | 33 |
| | TCR4-3β-30 | ASS**LTR**GLAGGIYEQY | 33 |
| | TCR4-3β-31 | ASS**LTH**GLAGGIYEQY | 186 |
| | TCR4-3β-32 | ASS**LTT**GLAGGIYEQY | 187 |
| | TCR4-3β-33 | ASS**LTS**GLAGGIYEQY | 34 |

Note:
the nucleotide sequence corresponding to * is Amber Stop Codon TAG

3) ELISA Assay of Phage

**[0101]** For the successfully screened CDRs, single clones from the small plates spread with dilution were selected and inoculated into 150 mL 2×YT medium (containing 100 μg/mL ampicillin and 2% glucose) and cultured at 37° C., 200-220 rpm for overnight. 2-5 μL of bacteria solution of each clone was pipetted into a new 96-well plate containing 200 μL of the same medium and cultured at 37° C., 200-220 rpm for 3 hours. Then 50 μL of helper phage with sufficient titer was added into each well and incubated at 37° C., 200-220 rpm for 1 hour.

**[0102]** The supernatant was discarded after the mixture was centrifuged with low speed for 10 minutes at room temperature, and the pellets were resuspended with 200 μL medium (2×YT medium with 100 μg/mL ampicillin+50 μg/mL kanamycin), and cultured at 26° C., 200-220 rpm for overnight.

**[0103]** 0.5 g of streptavidin was added to each well of a 96-well ELISA plate. The ELISA plate was kept at 4° C. in the refrigerator in stationary for overnight, then washed twice with PBS. 0.5 g of biotin-labeled pMHC was added to each well, and reacted for 30 minutes at room temperature. 400 μL of 3% milk PBS solution was added for blocking at room temperature for 1 hour.

**[0104]** After 100 μL of fresh phage supernatant and an identical volume of milk solution were added for 1 hour blocking at room temperature, 100 μL of the solution was collected and added to the ELSA plate which was washed with PBS for 3 rounds. The reaction was continued at room temperature for 1 hour.

**[0105]** The ELISA plate was washed with 0.1% PBST for 3 rounds, 100 μL of diluted M13-HRP antibody (Sinobiological, Cat. No. 11973-MM05T-H, with a dilution ratio of 1:10000) was added, and reacted for 30 minutes at room temperature. The ELISA plate was washed again with 0.1% PBST for 5 rounds, and 50 μL color development solution was added to each well to develop color for 90 seconds. Then the reaction was terminated immediately, followed by the determination of absorption value ($OD_{450}$) via the microplate reader.

**[0106]** The result of TCR2 is taken as an example for phage ELISA assay. FIG. **4** is the photograph of the ELISA plate after the 2nd round screening of single clone ELISA assay for TCR2, and Table 7 shows the $OD_{450}$ readings of the plate by the microplate reader.

TABLE 7

| Readings of ELISA | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| A | 0.389 | 0.032 | 0.042 | 0.302 | 0.035 | 0.036 | 0.369 | 0.323 | 0.043 | 0.042 | 0.051 | 0.054 |
| B | 0.042 | 0.525 | 0.230 | 0.038 | 0.032 | 0.033 | 0.400 | 0.511 | 0.498 | 0.446 | 0.373 | 0.506 |
| C | 0.292 | 0.042 | 0.037 | 0.034 | 0.475 | 0.466 | 0.031 | 0.186 | 0.219 | 0.042 | 0.382 | 0.047 |
| D | 0.044 | 0.545 | 0.476 | 0.037 | 0.037 | 0.036 | 0.341 | 0.033 | 0.435 | 0.607 | 0.582 | 0.822 |
| E | 0.468 | 0.037 | 0.071 | 0.037 | 0.503 | 0.516 | 0.036 | 0.466 | 0.042 | 0.279 | 0.517 | 0.122 |
| F | 0.043 | 0.364 | 0.524 | 0.461 | 0.252 | 0.525 | 0.036 | 0.041 | 0.560 | 0.041 | 0.485 | 0.710 |
| G | 0.036 | 0.051 | 0.457 | 0.037 | 0.039 | 0.034 | 0.274 | 0.035 | 0.184 | 0.038 | 0.045 | 0.090 |
| H | 0.039 | 0.496 | 0.066 | 0.426 | 0.046 | 0.039 | 0.037 | 0.097 | 0.377 | 0.047 | 0.052 | 0.038 |

**[0107]** The single clones with $OD_{450}$ values greater than 0.2 in Table 7 were selected and sent to the sequencing company for sequencing analysis. The corresponding mutant sequences were obtained according to the sequencing results (see Table 8).

TABLE 8

| ELISA sequencing results of TCR2 CDR3α | | | |
|---|---|---|---|
| Clone Number | CDR3α$_{(G90-T97)}$ | Sequence Frequency | SEQ ID NO: |
| TCR2-3α-wt | GGGADGLT | | |
| TCR2-3α-ELISA-1 | GIGDDGLT | 1 | 40 |
| TCR2-3α-ELISA-2 | GAGTDGLT | 1 | 41 |
| TCR2-3α-ELISA-3 | GAGVDGLT | 1 | 42 |
| TCR2-3α-ELISA-4 | GVGVDGLT | 4 | 43 |
| TCR2-3α-ELISA-5 | GVGDDGLT | 1 | 44 |
| TCR2-3α-ELISA-6 | AGGVDGLT | 4 | 45 |
| TCR2-3α-ELISA-7 | AAGVDGLT | 8 | 46 |
| TCR2-3α-ELISA-8 | AAGTDGLT | 2 | 47 |
| TCR2-3α-ELISA-9 | AVGDDGLT | 3 | 48 |
| TCR2-3α-ELISA-10 | AIGDDGLT | 2 | 49 |
| TCR2-3α-ELISA-11 | AEGVDGLT | 2 | 50 |
| TCR2-3α-ELISA-12 | SVGDDGLT | 1 | 51 |
| TCR2-3α-ELISA-13 | AVGVDGLT | 1 | 52 |
| TCR2-3α-ELISA-14 | AVGTDGLT | 1 | 53 |

**[0108]** obtained mutant sequences are listed in Table 9 ("A" represents α chain, "B" represents β chain, and the numbers with different "A" or "B" indicate the "α chain" or "β chain" containing different mutations; additionally, the heterodimer consisting of TCR α chain and TCR β chain is abbreviated as AmBn in the following examples; wherein m is an integer from 0 to N and n is an integer from 0 to N).

TABLE 9

| Mutant sequences of TCR1, TCR3 and TCR4 | | | | |
|---|---|---|---|---|
| TCR | Sequence Number | CDR3α | CDR3β | SEQ ID NO: |
| TCR1 | A0 | AVSNSGAGSYQLT | | 18 |
| | B0 | | ASSLGLGEETQY | 20 |

TABLE 9-continued

Mutant sequences of TCR1, TCR3 and TCR4

| TCR | Sequence Number | CDR3α | CDR3β | SEQ ID NO: |
|---|---|---|---|---|
| | A1 | AVSNSLDGYYQLT | | 19 |
| | B1 | | ASSLGYGRETQY | 21 |
| | B2 | | ASSLGFGRETQY | 22 |
| | B3 | | ASSLGYGTETQY | 23 |
| | B4 | | ASSLGYGQETQY | 24 |
| TCR3 | B0 | | ASRHLYNEQF | 25 |
| | B1 | | ASQSWYNEQF | 26 |
| | B2 | | ASQRWYNEQF | 27 |
| | B3 | | ASRREFNEQF | 28 |
| | B4 | | ASRHEFDEQF | 29 |
| | B5 | | ASRLLFNEQF | 30 |
| TCR4 | B0 | | ASSQGIGLAGGIYEQY | 31 |
| | B1 | | ASSLTIGLAGGIYEQY | 32 |
| | B2 | | ASSLTRGLAGGIYEQY | 33 |
| | B3 | | ASSLTSGLAGGIYEQY | 34 |
| | B4 | | ASALTWGLAGGIYEQY | 35 |
| | B5 | | ASSLGIGLAGGIYEQY | 36 |
| | B6 | | ASSQTIGLAGGIYEQY | 37 |
| | B7 | | ASSQGRGLAGGIYEQY | 38 |
| | B8 | | ASSQGSGLAGGIYEQY | 39 |

## Example 5: Inclusion Body Expression, TCR In Vitro Refolding and Affinity Test

### 1) Inclusion Body Expression

**[0109]** Some of the mutant sequences obtained in Example 3 and Example 4 were selected, whose amino acid mutations were transplanted into A0 or B0 of the TCR. The mutants were transformed into *E. coli* BL21 (DE3) respectively, and single clones were selected and inoculated into LB medium, cultured at 37° C., 250 rpm with shaking until $OD_{600}$=0.6-0.8. IPTG was then added to a final concentration of 0.8 mM and the bacteria were continued to be cultured at 37° C. for 3 h. The bacteria pellets were collected by centrifugation at 6000 rpm for 10 min and stored at −20° C.

**[0110]** The bacteria pellets were resuspended with lysis solution (PBS containing 0.5% TritonX-100), extracted by ultrasonication, and then centrifuged at a high speed of 12000 rpm for 20 min. The supernatant was discarded, and the precipitates were resuspended with lysis solution until no pellet was visible, then were subject to centrifugation at high speed for 10 min. The above operation was repeated for 2-3 rounds. The precipitates were dissolved with 6 M guanidine hydrochloride solution, centrifuged at high speed for 10 min to collect the supernatant, which referred as the purified inclusion body. 1 μL of the supernatant was subject to perform SDS-PAGE to identify the purity of inclusion body. The inclusion body was quantified, separated, and froze for storage at −80° C.

### 2) TCR In Vitro Refolding

**[0111]** 15 mg inclusion bodies of TCR α chain and 10 mg inclusion bodies TCR β chain were diluted in 5 mL 6M guanidine hydrochloride solution, respectively. The TCR α chain and TCR β chain were slowly added sequentially to the pre-cooled refolding buffer (Science 1996, 274, 209; *J. Mol. Biol.* 1999, 285, 1831; *Protein Eng.* 2003, 16, 707) and mixed with continuous stirring for 30 min at 4° C. The mixture was then put into the dialysis bag and dialyzed in 10 times the volume of pre-cooled deionized water with stirring for 8-12 hours. The dialysis was performed in pre-cooled dialysis solution (pH 8.1, 20 mM Tris-HCl) at 4° C. for 8 hours, and repeated for 2-3 rounds.

**[0112]** The solution was poured out from dialysis bag and centrifuged at high speed for 10 min to remove precipitation and air bubbles, and subject to anion exchange chromatography with HiTrap Q HP (5 mL). The solution was linearly eluted with 0-2 M NaCl, 20 mM Tris, pH 8.1. The elution peaks were collected for non-reducing SDS-PAGE, and the elution peaks containing the target protein fractions were combined and concentrated. The concentrated protein samples were subject to size-exclusion chromatography with superdex 75 10/300. Taking A0B1 of TCR3 as an example, the results of protein purification and SDS-PAGE are shown in FIG. **5**A, FIG. **5**B, FIG. **6**A and FIG. **6**B.

### 3) Affinity Test

**[0113]** Biacore is an instrument for affinity based using surface plasmon resonance (SPR) technique. The binding and dissociation constants were calculated by monitoring the change of SPR angle and analysis. In this experiment, biotinylated pMHC was coupled on the biosensor surface and its binding and dissociation constants with different TCRs were detected to calculate the $K_D$ values. Biacore T200 was used to perform the affinity test for the purified TCR. Taking A0B4 of TCR1 as an example, its affinity with HLA-A*02:01/WT1 (peptide RMFPNAPYL) was tested, the details are shown in FIG. **7**.

**[0114]** Table 10 summarizes the affinity between each refolding-validated TCR mutant and its corresponding pMHC. It can be seen that the mutants obtained by screening with the designed method for heterodimeric display method have more than 2-fold improvement in affinity compared with the wild-type TCR.

TABLE 10

Affinity of some mutants

| TCR | Validation Number | CDR3α | SEQ ID NO: | CDR3β | SEQ ID NO: | $K_D$(M) |
|---|---|---|---|---|---|---|
| TCR1 | A0B0 | AVSNSGAGSYQLT | 18 | ASSLGLGEETQY | 20 | 5.5E-05 |
| | A0B3 | AVSNSGAGSYQLT | 18 | ASSLGYGTETQY | 23 | 6.5E-08 |
| | A0B4 | AVSNSGAGSYQLT | 18 | ASSLGYGQETQY | 24 | 3.5E-07 |
| TCR2 | A0B0 | GGGADGLT | 118 | ASSLYGGGDT | 189 | 6.9E-05 |
| | A8B0 | AEGVDGLT | 50 | ASSLYGGGDT | 189 | 1.5E-05 |
| | A9B0 | GVGDDGLT | 44 | ASSLYGGGDT | 189 | 2.5E-05 |
| TCR3 | A0B0 | AVEGDSNYQLI | 188 | ASRHLYNEQF | 25 | 6.1E-05 |
| | A0B1 | AVEGDSNYQLI | 188 | ASQSWYNEQF | 26 | 1.7E-05 |
| | A0B2 | AVEGDSNYQLI | 188 | ASQRWYNEQF | 27 | 1.5E-05 |
| | A0B3 | AVEGDSNYQLI | 188 | ASRREFNEQF | 28 | 1.9E-05 |
| | A0B4 | AVEGDSNYQLI | 188 | ASRHEFDEQF | 29 | 1.3E-05 |
| | A0B5 | AVEGDSNYQLI | 188 | ASRLLFNEQF | 30 | 3.1E-05 |

**[0115]** The present invention creatively places A1 peptide coding sequence at the 3' end of TCR α chain coding sequence to encode fusion protein: α chain-A1 peptide; places B1 peptide coding sequence at the 3' end of TCR β chain coding sequence (expression with gene III, with Amber Stop Codon between the two, by fusion) to encode fusion protein: β chain-B1 peptide-pIII; the expression level of α chain-A1 peptide fusion protein is much higher than that of R chain-B1 peptide-pIII fusion protein. The expression level of B1 peptide is extremely low, reducing the possibility of forming a β chain-B1 peptide-pIII fusion protein homodimer, and allowing more TCR monovalent display on the phage surface at the same time. If the α chain-A1 peptide forms a homodimer due to the formation of A1 peptide dimer, the dimer does not link to phage pIII protein and therefore cannot display on phage surface. The αβ heterodimeric TCR forms only when A1 and B1 form a heterodimer to promote the pairing of α chain and β chain of the TCR. skp molecular chaperone further promotes the folding of both chains of the TCR to improve the formation rate of the αβ heterodimeric TCR. After α chain-A1 peptide and β chain-B1 peptide form the αβ heterodimeric TCR, it will display on the phage surface by the pIII protein expressed by fusion. The present invention utilizes A1 peptide and B1 peptide, with the assistance of skp, to promote the correct pairing and folding of αβ heterodimeric TCR, which in turn enhances the display efficiency of correctly folded TCRs on the phage surface. The experimental results show that the present invention may efficiently accomplish the screening of αβ heterodimeric TCR mutants. The method may also be used to optimize the stability, water solubility, and affinity of TCRs.

**[0116]** Although the above describes specific embodiments of the present invention, it should be understood by those skilled in the art that these are merely illustrative examples and that a variety of changes or modifications can be made to these embodiments without departing from the principles and substance of the present invention. Therefore, the scope of protection of the present disclosure is limited by the appended claims.

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<223> OTHER INFORMATION: A1 peptide

<400> SEQUENCE: 1

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Gln Asn Tyr Glu Leu Lys Ser Arg Val Gln Arg Leu Arg Glu Gln Val
20 25 30

Ala Gln Leu Ala Ser
35

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<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: B1 peptide

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Ser Thr Ser Val Asp Glu Leu Gln Ala Glu Val Asp Gln Leu Gln Asp
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Glu Asn Tyr Ala Leu Lys Thr Lys Val Ala Gln Leu Arg Lys Lys Val
            20                  25                  30

Glu Lys Leu Ser Glu
        35

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<223> OTHER INFORMATION: linker

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Gly Gly Ser Gly Gly Gly Gly Gly
1               5

<210> SEQ ID NO 4
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<220> FEATURE:
<223> OTHER INFORMATION: chain constant region

<400> SEQUENCE: 4

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Ser Glu Ala Glu Ile Ser His Thr Gln Lys Ala Thr Leu Val Cys Leu
            20                  25                  30

Ala Thr Gly Phe Phe Pro Asp His Val Glu Leu Ser Trp Trp Val Asn
        35                  40                  45

Gly Lys Glu Val His Ser Gly Val Ser Thr Asp Pro Gln Pro Leu Lys
    50                  55                  60

Glu Gln Pro Ala Leu Asn Asp Ser Arg Tyr Cys Leu Ser Ser Arg Leu
65                  70                  75                  80

Arg Val Ser Ala Thr Phe Trp Gln Asn Pro Arg Asn His Phe Arg Cys
                85                  90                  95

Gln Val Gln Phe Tyr Gly Leu Ser Glu Asn Asp Glu Trp Thr Gln Asp
            100                 105                 110

Arg Ala Lys Pro Val Thr Gln Ile Val Ser Ala Glu Ala Trp Gly Arg
        115                 120                 125

Ala Asp
    130

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-continued

```
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            20                  25                  30

Val Ser Gln Ser Lys Asp Ser Asp Val Tyr Ile Thr Asp Lys Thr Val
        35                  40                  45

Leu Asp Met Arg Ser Met Asp Phe Lys Ser Asn Ser Ala Val Ala Trp
    50                  55                  60

Ser Asn Lys Ser Asp Phe Ala Cys Ala Asn Ala Phe Asn Asn Ser Ile
65                  70                  75                  80

Ile Pro Glu Asp Thr Phe Phe Pro Ser Pro Glu Ser Ser
                85                  90
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            20                  25                  30

Ala Thr Gly Phe Tyr Pro Asp His Val Glu Leu Ser Trp Trp Val Asn
        35                  40                  45

Gly Lys Glu Val His Ser Gly Val Ser Thr Asp Pro Gln Pro Leu Lys
    50                  55                  60

Glu Gln Pro Ala Leu Asn Asp Ser Arg Tyr Cys Leu Ser Ser Arg Leu
65                  70                  75                  80

Arg Val Ser Ala Thr Phe Trp Gln Asn Pro Arg Asn His Phe Arg Cys
                85                  90                  95

Gln Val Gln Phe Tyr Gly Leu Ser Glu Asn Asp Glu Trp Thr Gln Asp
            100                 105                 110

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Ala Asp
    130
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ggtgtttcta acacgctgga aaatgagttc aaaggccgtg ccagcgaact gcagcgtatg    180

gaaaccgatc tgcaggctaa aatgaaaaag ctgcagtcca tgaaagcggg cagcgatcgc    240

actaagctgg aaaaagacgt gatggctcag cgccagactt ttgctcagaa agcgcaggct    300

tttgagcagg atcgcgcacg tcgttccaac gaagaacgcg gcaaactggt tactcgtatc    360

cagactgctg tgaaatccgt tgccaacagc caggatatcg atctggttgt tgatgcaaac    420

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<223> OTHER INFORMATION: between the signal peptide and the Amber Stop
      Codon

<400> SEQUENCE: 9

atgaaatacc tattgcctac ggcagccgct ggattgttat tactcgcggc ccagccggcc     60

atggctctaa tagtgactaa tagtgactaa tagtgagtga gcaagggcga ggagctgttc    120

accggggtgg tgcccatcct ggtcgagctg gacggcgacg taaacggcca caagttcagc    180

gtgtccggcg agggcgaggg cgatgccacc tacggcaagc tgaccctgaa gttcatctgc    240

accaccggca agctgcccgt gccctggccc accctcgtga ccaccctgac ctacggcgtg    300

cagtgcttca gccgctaccc cgaccacatg aagcagcacg acttcttcaa gtccgccatg    360

cccgaaggct acgtccagga gcgcaccatc ttcttcaagg acgacggcaa ctacaagacc    420

cgcgccgagg tgaagttcga gggcgacacc ctggtgaacc gcatcgagct gaagggcatc    480

gacttcaagg aggacggcaa catcctgggg cacaagctgg agtacaacta caacagccac    540

aacgtctata tcatggccga caagcagaag aacggcatca aggtgaactt caagatccgc    600

cacaacatcg aggacggcag cgtgcagctc gccgaccact accagcagaa cacccccatc    660

ggcgacggcc ccgtgctgct gcccgacaac cactacctga gcacccagtc cgccctgagc    720

aaagacccca acgagaagcg cgatcacatg gtcctgctgg agttcgtgac cgccgccggg    780

atcactctcg gcatggacga gctgtacaag taatagtgac taatagtgac taatagtgat    840

gcggccgctg gatccgaaca gaaactgatc agcgaagagg atctggctga atag          894
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<210> SEQ ID NO 10
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild-type TCR1

<400> SEQUENCE: 10

Gly Gln Asn Ile Asp Gln Pro Thr Glu Met Thr Ala Thr Glu Gly Ala
1               5                   10                  15

Ile Val Gln Ile Asn Cys Thr Tyr Gln Thr Ser Gly Phe Asn Gly Leu
            20                  25                  30

Phe Trp Tyr Gln Gln His Ala Gly Glu Ala Pro Thr Phe Leu Ser Tyr
```

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```
        35                  40                  45

Asn Val Leu Asp Gly Leu Glu Glu Lys Gly Arg Phe Ser Ser Phe Leu
    50                  55                  60

Ser Arg Ser Lys Gly Tyr Ser Tyr Leu Leu Leu Lys Glu Leu Gln Met
65                  70                  75                  80

Lys Asp Ser Ala Ser Tyr Leu Cys Ala Val Ser Asn Ser Gly Ala Gly
                85                  90                  95

Ser Tyr Gln Leu Thr Phe Gly Lys Gly Thr Lys Leu Ser Val Ile Pro
            100                 105                 110

Asn


<210> SEQ ID NO 11
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild-type TCR1

<400> SEQUENCE: 11

Asp Thr Gly Val Ser Gln Asp Pro Arg His Lys Ile Thr Lys Arg Gly
1               5                   10                  15

Gln Asn Val Thr Phe Arg Cys Asp Pro Ile Ser Glu His Asn Arg Leu
            20                  25                  30

Tyr Trp Tyr Arg Gln Thr Leu Gly Gln Gly Pro Glu Phe Leu Thr Tyr
        35                  40                  45

Phe Gln Asn Glu Ala Gln Leu Glu Lys Ser Arg Leu Leu Ser Asp Arg
    50                  55                  60

Phe Ser Ala Glu Arg Pro Lys Gly Ser Phe Ser Thr Leu Glu Ile Gln
65                  70                  75                  80

Arg Thr Glu Gln Gly Asp Ser Ala Met Tyr Leu Cys Ala Ser Ser Leu
                85                  90                  95

Gly Leu Gly Glu Glu Thr Gln Tyr Phe Gly Pro Gly Thr Arg Leu Leu
            100                 105                 110

Val Leu


<210> SEQ ID NO 12
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild-tpye TCR2

<400> SEQUENCE: 12

Glu Leu Lys Val Glu Gln Asn Pro Leu Phe Leu Ser Met Gln Glu Gly
1               5                   10                  15

Lys Asn Tyr Thr Ile Tyr Cys Asn Tyr Ser Thr Thr Ser Asp Arg Leu
            20                  25                  30

Tyr Trp Tyr Arg Gln Asp Pro Gly Lys Ser Leu Glu Ser Leu Phe Val
        35                  40                  45

Leu Leu Ser Asn Gly Ala Val Lys Gln Glu Gly Arg Leu Met Ala Ser
    50                  55                  60

Leu Asp Thr Lys Ala Arg Leu Ser Thr Leu His Ile Thr Ala Ala Val
65                  70                  75                  80

His Asp Leu Ser Ala Thr Tyr Phe Cys Gly Gly Gly Ala Asp Gly Leu
                85                  90                  95
```

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```
Thr Phe Gly Lys Gly Thr His Leu Ile Ile Gln Pro Tyr
            100                 105


<210> SEQ ID NO 13
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild-tpye TCR2

<400> SEQUENCE: 13

Asp Thr Gly Val Ser Gln Asp Pro Arg His Lys Ile Thr Lys Arg Gly
1               5                   10                  15

Gln Asn Val Thr Phe Arg Cys Asp Pro Ile Ser Glu His Asn Arg Leu
            20                  25                  30

Tyr Trp Tyr Arg Gln Thr Leu Gly Gln Gly Pro Glu Phe Leu Thr Tyr
        35                  40                  45

Phe Gln Asn Glu Ala Gln Leu Glu Lys Ser Arg Leu Leu Ser Asp Arg
    50                  55                  60

Phe Ser Ala Glu Arg Pro Lys Gly Ser Phe Ser Thr Leu Glu Ile Gln
65                  70                  75                  80

Arg Thr Glu Gln Gly Asp Ser Ala Met Tyr Leu Cys Ala Ser Ser Leu
                85                  90                  95

Tyr Gly Gly Gly Asp Thr Gln Tyr Phe Gly Pro Gly Thr Arg Leu Thr
            100                 105                 110

Val Leu


<210> SEQ ID NO 14
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild-type TCR3

<400> SEQUENCE: 14

Gly Ile Gln Val Glu Gln Ser Pro Pro Asp Leu Ile Leu Gln Glu Gly
1               5                   10                  15

Ala Asn Ser Thr Leu Arg Cys Asn Phe Ser Asp Ser Val Asn Asn Leu
            20                  25                  30

Gln Trp Phe His Gln Asn Pro Trp Gly Gln Leu Ile Asn Leu Phe Tyr
        35                  40                  45

Ile Pro Ser Gly Thr Lys Gln Asn Gly Arg Leu Ser Ala Thr Thr Val
    50                  55                  60

Ala Thr Glu Arg Tyr Ser Leu Leu Tyr Ile Ser Ser Ser Gln Thr Thr
65                  70                  75                  80

Asp Ser Gly Val Tyr Phe Cys Ala Val Glu Gly Asp Ser Asn Tyr Gln
                85                  90                  95

Leu Ile Trp Gly Ala Gly Thr Lys Leu Ile Ile Lys Pro Asp
            100                 105                 110


<210> SEQ ID NO 15
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild-type TCR3
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<400> SEQUENCE: 15

```
Asp Thr Gly Val Ser Gln Asp Pro Arg His Lys Ile Thr Lys Arg Gly
1               5                   10                  15

Gln Asn Val Thr Phe Arg Cys Asp Pro Ile Ser Glu His Asn Arg Leu
            20                  25                  30

Tyr Trp Tyr Arg Gln Thr Leu Gly Gln Gly Pro Glu Phe Leu Thr Tyr
        35                  40                  45

Phe Gln Asn Glu Ala Gln Leu Glu Lys Ser Arg Leu Leu Ser Asp Arg
    50                  55                  60

Phe Ser Ala Glu Arg Pro Lys Gly Ser Phe Ser Thr Leu Glu Ile Gln
65                  70                  75                  80

Arg Thr Glu Gln Gly Asp Ser Ala Met Tyr Leu Cys Ala Ser Arg His
                85                  90                  95

Leu Tyr Asn Glu Gln Phe Phe Gly Pro Gly Thr Arg Leu Thr Val Leu
            100                 105                 110
```

<210> SEQ ID NO 16
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild-type TCR4

<400> SEQUENCE: 16

```
Ala Gln Lys Ile Thr Gln Thr Gln Pro Gly Met Phe Val Gln Glu Lys
1               5                   10                  15

Glu Ala Val Thr Leu Asp Cys Thr Tyr Asp Thr Ser Asp Gln Ser Tyr
            20                  25                  30

Gly Leu Phe Trp Tyr Lys Gln Pro Ser Ser Gly Glu Met Ile Phe Leu
        35                  40                  45

Ile Tyr Gln Gly Ser Tyr Asp Glu Gln Asn Ala Thr Glu Gly Arg Tyr
    50                  55                  60

Ser Leu Asn Phe Gln Lys Ala Arg Lys Ser Ala Asn Leu Val Ile Ser
65                  70                  75                  80

Ala Ser Gln Leu Gly Asp Ser Ala Met Tyr Phe Cys Ala Met Arg Glu
                85                  90                  95

Pro Arg Gly Ser Ala Arg Gln Leu Thr Phe Gly Ser Gly Thr Gln Leu
            100                 105                 110

Thr Val Leu Pro Asp
        115
```

<210> SEQ ID NO 17
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: wild-type TCR4

<400> SEQUENCE: 17

```
Asp Gly Gly Ile Thr Gln Ser Pro Lys Tyr Leu Phe Arg Lys Glu Gly
1               5                   10                  15

Gln Asn Val Thr Leu Ser Cys Glu Gln Asn Leu Asn His Asp Ala Met
            20                  25                  30

Tyr Trp Tyr Arg Gln Asp Pro Gly Gln Gly Leu Arg Leu Ile Tyr Tyr
        35                  40                  45
```

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```
Ser Gln Ile Val Asn Asp Phe Gln Lys Gly Asp Ile Ala Glu Gly Tyr
    50                  55                  60

Ser Val Ser Arg Glu Lys Lys Glu Ser Phe Pro Leu Thr Val Thr Ser
65                  70                  75                  80

Ala Gln Lys Asn Pro Thr Ala Phe Tyr Leu Cys Ala Ser Ser Gln Gly
                85                  90                  95

Ile Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr Phe Gly Pro Gly Thr
            100                 105                 110

Arg Leu Thr Val Thr
        115


<210> SEQ ID NO 18
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1 CDR3 A0

<400> SEQUENCE: 18

Ala Val Ser Asn Ser Gly Ala Gly Ser Tyr Gln Leu Thr
1               5                   10


<210> SEQ ID NO 19
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1 CDR3 A1

<400> SEQUENCE: 19

Ala Val Ser Asn Ser Leu Asp Gly Tyr Tyr Gln Leu Thr
1               5                   10


<210> SEQ ID NO 20
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1 CDR3 B0

<400> SEQUENCE: 20

Ala Ser Ser Leu Gly Leu Gly Glu Glu Thr Gln Tyr
1               5                   10


<210> SEQ ID NO 21
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1 CDR3 B1

<400> SEQUENCE: 21

Ala Ser Ser Leu Gly Tyr Gly Arg Glu Thr Gln Tyr
1               5                   10


<210> SEQ ID NO 22
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1 CDR3 B2

<400> SEQUENCE: 22
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```
Ala Ser Ser Leu Gly Phe Gly Arg Glu Thr Gln Tyr
1               5                   10


<210> SEQ ID NO 23
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1 CDR3 B3

<400> SEQUENCE: 23

Ala Ser Ser Leu Gly Tyr Gly Thr Glu Thr Gln Tyr
1               5                   10


<210> SEQ ID NO 24
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1 CDR3 B4

<400> SEQUENCE: 24

Ala Ser Ser Leu Gly Tyr Gly Gln Glu Thr Gln Tyr
1               5                   10


<210> SEQ ID NO 25
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TR3 CDR3 B0

<400> SEQUENCE: 25

Ala Ser Arg His Leu Tyr Asn Glu Gln Phe
1               5                   10


<210> SEQ ID NO 26
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3 CDR3 B1

<400> SEQUENCE: 26

Ala Ser Gln Ser Trp Tyr Asn Glu Gln Phe
1               5                   10


<210> SEQ ID NO 27
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3 CDR3 B2

<400> SEQUENCE: 27

Ala Ser Gln Arg Trp Tyr Asn Glu Gln Phe
1               5                   10


<210> SEQ ID NO 28
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3 CDR3 B3
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<400> SEQUENCE: 28

Ala Ser Arg Arg Glu Phe Asn Glu Gln Phe
1               5                   10


<210> SEQ ID NO 29
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3 CDR3 B4

<400> SEQUENCE: 29

Ala Ser Arg His Glu Phe Asp Glu Gln Phe
1               5                   10


<210> SEQ ID NO 30
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3 CDR3 B5

<400> SEQUENCE: 30

Ala Ser Arg Leu Leu Phe Asn Glu Gln Phe
1               5                   10


<210> SEQ ID NO 31
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4 CDR3 B0

<400> SEQUENCE: 31

Ala Ser Ser Gln Gly Ile Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15


<210> SEQ ID NO 32
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4 CDR3 B1

<400> SEQUENCE: 32

Ala Ser Ser Leu Thr Ile Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15


<210> SEQ ID NO 33
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4 CDR3 B2

<400> SEQUENCE: 33

Ala Ser Ser Leu Thr Arg Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15


<210> SEQ ID NO 34
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
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<220> FEATURE:
<223> OTHER INFORMATION: TCR4 CDR3 B3

<400> SEQUENCE: 34

Ala Ser Ser Leu Thr Ser Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 35
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4 CDR3 B4

<400> SEQUENCE: 35

Ala Ser Ala Leu Thr Trp Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 36
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4 CDR3 B5

<400> SEQUENCE: 36

Ala Ser Ser Leu Gly Ile Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 37
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4 CDR3 B6

<400> SEQUENCE: 37

Ala Ser Ser Gln Thr Ile Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 38
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4 CDR3 B7

<400> SEQUENCE: 38

Ala Ser Ser Gln Gly Arg Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 39
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4 CDR3 B8

<400> SEQUENCE: 39

Ala Ser Ser Gln Gly Ser Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 40
<211> LENGTH: 8

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-1

<400> SEQUENCE: 40

Gly Ile Gly Asp Asp Gly Leu Thr
1               5

<210> SEQ ID NO 41
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-2

<400> SEQUENCE: 41

Gly Ala Gly Thr Asp Gly Leu Thr
1               5

<210> SEQ ID NO 42
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-3

<400> SEQUENCE: 42

Gly Ala Gly Val Asp Gly Leu Thr
1               5

<210> SEQ ID NO 43
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-4

<400> SEQUENCE: 43

Gly Val Gly Val Asp Gly Leu Thr
1               5

<210> SEQ ID NO 44
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-5

<400> SEQUENCE: 44

Gly Val Gly Asp Asp Gly Leu Thr
1               5

<210> SEQ ID NO 45
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-6

<400> SEQUENCE: 45

Ala Gly Gly Val Asp Gly Leu Thr
1               5

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<210> SEQ ID NO 46
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-7

<400> SEQUENCE: 46

Ala Ala Gly Val Asp Gly Leu Thr
1               5

<210> SEQ ID NO 47
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-8

<400> SEQUENCE: 47

Ala Ala Gly Thr Asp Gly Leu Thr
1               5

<210> SEQ ID NO 48
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-9

<400> SEQUENCE: 48

Ala Val Gly Asp Asp Gly Leu Thr
1               5

<210> SEQ ID NO 49
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-10

<400> SEQUENCE: 49

Ala Ile Gly Asp Asp Gly Leu Thr
1               5

<210> SEQ ID NO 50
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-11

<400> SEQUENCE: 50

Ala Glu Gly Val Asp Gly Leu Thr
1               5

<210> SEQ ID NO 51
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-12

<400> SEQUENCE: 51

Ser Val Gly Asp Asp Gly Leu Thr
1               5

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<210> SEQ ID NO 52
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-13

<400> SEQUENCE: 52

Ala Val Gly Val Asp Gly Leu Thr
1               5

<210> SEQ ID NO 53
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3-ELISA-14

<400> SEQUENCE: 53

Ala Val Gly Thr Asp Gly Leu Thr
1               5

<210> SEQ ID NO 54
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: WT1

<400> SEQUENCE: 54

Arg Met Phe Pro Asn Ala Pro Tyr Leu
1               5

<210> SEQ ID NO 55
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AFP

<400> SEQUENCE: 55

Phe Met Asn Lys Phe Ile Tyr Glu Ile
1               5

<210> SEQ ID NO 56
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: PRAME

<400> SEQUENCE: 56

Val Leu Asp Gly Leu Asp Val Leu Leu
1               5

<210> SEQ ID NO 57
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pPD-F

<400> SEQUENCE: 57

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acacaggaaa cagctatgac 20

<210> SEQ ID NO 58
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pPD-R

<400> SEQUENCE: 58

acactgagtt tcgtcaccag 20

<210> SEQ ID NO 59
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3alpha-F1

<400> SEQUENCE: 59

ggttcctacc agctgacctt cg 22

<210> SEQ ID NO 60
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3alpha-R1
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (27)..(28)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (30)..(31)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (33)..(34)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (36)..(37)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (39)..(40)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 60

aggtcagctg gtaggaacca gcaccmnnmn nmnnmnnmnn gcacaggtag gacgcag 57

<210> SEQ ID NO 61
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3alpha-R2
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (21)..(22)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (24)..(25)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (27)..(28)

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<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (30)..(31)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (33)..(34)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 61

aggtcagctg gtaggaaccm nnmnnmnnmn nmnntaccgc gcacaggtag gac 53

<210> SEQ ID NO 62
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3alpha-F2

<400> SEQUENCE: 62

ttcggtaaag gtaccaaact g 21

<210> SEQ ID NO 63
<211> LENGTH: 59
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3alpha-R3
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (34)..(35)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (37)..(38)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (40)..(41)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (43)..(44)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 63

agtttggtac ctttaccgaa ggtcagctgg tamnnmnnmn nmnnggagtt agataccgc 59

<210> SEQ ID NO 64
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3alpha-R4
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(29)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (31)..(32)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (34)..(35)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (37)..(38)

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<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 64

agtttggtac ctttaccgaa ggtcagmnnm nnmnnmnnag caccggagtt agatacc 57

<210> SEQ ID NO 65
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3alpha-R5
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (22)..(23)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (25)..(26)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (28)..(29)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (31)..(32)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 65

agtttggtac ctttaccgaa mnnmnnmnnm nnggaaccag caccggag 48

<210> SEQ ID NO 66
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3beta-R1

<400> SEQUENCE: 66

acacaggtac atcgcagaat cg 22

<210> SEQ ID NO 67
<211> LENGTH: 50
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3beta-F1
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (20)..(21)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (23)..(24)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (26)..(27)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (29)..(30)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 67

ttctgcgatg tacctgtgtn nknnknnknn kggtctgggt gaagaaaccc 50

<210> SEQ ID NO 68

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<211> LENGTH: 55
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3beta-F2
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (26)..(27)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (29)..(30)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (32)..(33)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (35)..(36)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 68

ttctgcgatg tacctgtgtg catctnnknn knnknnkggt gaagaaaccc agtac 55

<210> SEQ ID NO 69
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3beta-R2

<400> SEQUENCE: 69

cagggaagat gcacacaggt ac 22

<210> SEQ ID NO 70
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3beta-F3
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (21)..(22)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (24)..(25)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (27)..(28)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (30)..(31)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 70

acctgtgtgc atcttccctg nnknnknnkn nkgaaaccca gtactttggt c 51

<210> SEQ ID NO 71
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3beta-F4
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (27)..(28)

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<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (30)..(31)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (33)..(34)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (36)..(37)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 71

acctgtgtgc atcttccctg ggtctgnnkn nknnknnkca gtactttggt ccgg 54

<210> SEQ ID NO 72
<211> LENGTH: 59
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CDR3beta-F5
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (33)..(34)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (36)..(37)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (39)..(40)
<223> OTHER INFORMATION: n is a, c, g, or t
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (42)..(43)
<223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 72

acctgtgtgc atcttccctg ggtctgggtg aannknnknn knnktttggt ccgggtacc 59

<210> SEQ ID NO 73
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-1

<400> SEQUENCE: 73

Pro Asp Arg Pro Glu Gly Ala Gly Ser Tyr Gln Leu Thr
1               5                   10

<210> SEQ ID NO 74
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-2

<400> SEQUENCE: 74

Ala Val Pro Leu Gly Thr Leu Gly Ser Tyr Gln Leu Thr
1               5                   10

<210> SEQ ID NO 75
<211> LENGTH: 13
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-3

<400> SEQUENCE: 75

Ala Val Pro Lys Val Gly Thr Gly Ser Tyr Gln Leu Thr
1 5 10

<210> SEQ ID NO 76
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-4

<400> SEQUENCE: 76

Ala Val Ser Asn Ser Arg Gly Lys Asp Tyr Gln Leu Thr
1 5 10

<210> SEQ ID NO 77
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-5

<400> SEQUENCE: 77

Ala Val Ser Asn Ser Gly Ala Glu Trp Ala Ala Leu Thr
1 5 10

<210> SEQ ID NO 78
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-6

<400> SEQUENCE: 78

Ala Val Ser Asn Ser Gly Ala Gly Ser Ser Ser Arg Ser
1 5 10

<210> SEQ ID NO 79
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-7

<400> SEQUENCE: 79

Ala Val Ser Asn Ser Gly Ala Gly Ser Asn Val Asn Asp
1 5 10

<210> SEQ ID NO 80
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-8
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (2)..(2)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 80

Ser Xaa Gly Thr Lys Gly Ala Gly Ser Tyr Gln Leu Thr

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1 5 10

<210> SEQ ID NO 81
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-9

<400> SEQUENCE: 81

Ala Val Cys Phe Ala Met Pro Gly Ser Tyr Gln Leu Thr
1 5 10

<210> SEQ ID NO 82
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-10
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 82

Ala Val Xaa Ile Tyr His Thr Gly Ser Tyr Gln Leu Thr
1 5 10

<210> SEQ ID NO 83
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-11

<400> SEQUENCE: 83

Ala Val Asp Asp Leu Pro Asn Gly Ser Tyr Gln Leu Thr
1 5 10

<210> SEQ ID NO 84
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-12

<400> SEQUENCE: 84

Ala Val Ser Asn Ser Phe Pro Ala Thr Tyr Gln Leu Thr
1 5 10

<210> SEQ ID NO 85
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-13

<400> SEQUENCE: 85

Ala Val Ser Asn Ser Leu Asn Leu Pro Tyr Gln Leu Thr
1 5 10

<210> SEQ ID NO 86
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-14

<400> SEQUENCE: 86

Ala Val Ser Asn Ser Leu Ala Pro Pro Tyr Gln Leu Thr
1               5                   10

<210> SEQ ID NO 87
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-15
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (8)..(9)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 87

Ala Val Ser Asn Ser Gly Arg Xaa Xaa Tyr Gln Leu Thr
1               5                   10

<210> SEQ ID NO 88
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-16

<400> SEQUENCE: 88

Ala Val Ser Asn Ser Gly Leu Pro Arg Tyr Gln Leu Thr
1               5                   10

<210> SEQ ID NO 89
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-17

<400> SEQUENCE: 89

Ala Val Ser Asn Ser Gly Ala Gly Ala Arg Thr Leu Thr
1               5                   10

<210> SEQ ID NO 90
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-18

<400> SEQUENCE: 90

Ala Val Ser Asn Ser Gly Ala Leu Asp His Ser Leu Thr
1               5                   10

<210> SEQ ID NO 91
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-19

<400> SEQUENCE: 91

Ala Val Ser Asn Ser Gly Ala Gln Val Ala Arg Leu Thr
1               5                   10

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<210> SEQ ID NO 92
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-20

<400> SEQUENCE: 92

Ala Val Ser Asn Ser Gly Gly Gly Ser Asn Gly Trp Ala
1               5                   10

<210> SEQ ID NO 93
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-21

<400> SEQUENCE: 93

Ala Val Arg Cys Thr Gln Pro Gly Ser Tyr Gln Leu Thr
1               5                   10

<210> SEQ ID NO 94
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-22

<400> SEQUENCE: 94

Ala Val Ser Asn Ser Phe Val Glu Leu Tyr Gln Leu Thr
1               5                   10

<210> SEQ ID NO 95
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-23

<400> SEQUENCE: 95

Ala Val Ser Asn Ser Gln Arg Val Leu Tyr Gln Leu Thr
1               5                   10

<210> SEQ ID NO 96
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-24
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (9)..(9)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 96

Ala Val Ser Asn Ser Gly Ala Ser Xaa Cys Val Leu Thr
1               5                   10

<210> SEQ ID NO 97
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: TCR1-3alpha-25

<400> SEQUENCE: 97

Ala Val Ser Asn Ser Gly Ala Tyr Val Phe Pro Leu Thr
1               5                   10

<210> SEQ ID NO 98
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3alpha-26

<400> SEQUENCE: 98

Ala Val Ser Asn Ser Gly Ala Gly Ser Arg Arg Trp Glu
1               5                   10

<210> SEQ ID NO 99
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-1

<400> SEQUENCE: 99

Pro Gln Glu Pro Gly Leu Gly Glu Glu Thr Gln Tyr
1               5                   10

<210> SEQ ID NO 100
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-2

<400> SEQUENCE: 100

Thr Asp Met Leu Gly Leu Gly Glu Glu Thr Gln Tyr
1               5                   10

<210> SEQ ID NO 101
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-3
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 101

Ala Pro Xaa Thr Gly Leu Gly Glu Glu Thr Gln Tyr
1               5                   10

<210> SEQ ID NO 102
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-4

<400> SEQUENCE: 102

Ala Ser Pro Arg Ala Leu Gly Glu Glu Thr Gln Tyr
1               5                   10

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<210> SEQ ID NO 103
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-5

<400> SEQUENCE: 103

Ala Ser Ser Leu Pro Thr Lys Glu Glu Thr Gln Tyr
1               5                   10


<210> SEQ ID NO 104
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-6

<400> SEQUENCE: 104

Ala Ser Ser Leu Gly Leu Leu Val Tyr Leu Gln Tyr
1               5                   10


<210> SEQ ID NO 105
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-7
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (8)..(8)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 105

Ala Ser Ser Leu Gly Leu Gln Xaa Leu Ala Gln Tyr
1               5                   10


<210> SEQ ID NO 106
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-8

<400> SEQUENCE: 106

Ala Ser Gly Met Val Ser Gly Glu Glu Thr Gln Tyr
1               5                   10


<210> SEQ ID NO 107
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-9
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 107

Ala Ser Xaa Thr Tyr Ala Gly Glu Glu Thr Gln Tyr
1               5                   10


<210> SEQ ID NO 108
<211> LENGTH: 12
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-10

<400> SEQUENCE: 108

Ala Ser Ser Leu Arg Glu Leu Ile Glu Thr Gln Tyr
1 5 10

<210> SEQ ID NO 109
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-11

<400> SEQUENCE: 109

Ala Ser Ser Leu Gly Met Gly Arg Glu Thr Gln Tyr
1 5 10

<210> SEQ ID NO 110
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-12

<400> SEQUENCE: 110

Ala Ser Ser Leu Gly Phe Gly His Glu Thr Gln Tyr
1 5 10

<210> SEQ ID NO 111
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-13

<400> SEQUENCE: 111

Ala Ser Ser Leu Gly Tyr Gly Ser Glu Thr Gln Tyr
1 5 10

<210> SEQ ID NO 112
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-15

<400> SEQUENCE: 112

Ala Ser Ser Leu Gly Leu Pro Pro Gly Ala Gln Tyr
1 5 10

<210> SEQ ID NO 113
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-16

<400> SEQUENCE: 113

Ala Ser Ser Leu Gly Leu Trp Arg Arg Arg Gln Tyr
1 5 10

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<210> SEQ ID NO 114
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-17

<400> SEQUENCE: 114

Ala Ser Ser Leu Gly Leu Gly Arg Glu Ile Gln Tyr
1               5                   10

<210> SEQ ID NO 115
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-18

<400> SEQUENCE: 115

Ala Ser Ser Leu Gly Leu Gly Tyr Asp Tyr Gln Tyr
1               5                   10

<210> SEQ ID NO 116
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-19

<400> SEQUENCE: 116

Ala Ser Ser Leu Gly Leu Gly Cys Leu Val Gln Tyr
1               5                   10

<210> SEQ ID NO 117
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR1-3beta-35

<400> SEQUENCE: 117

Ala Ser Ser Leu Gly Tyr Gly Ala Glu Thr Gln Tyr
1               5                   10

<210> SEQ ID NO 118
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-wt

<400> SEQUENCE: 118

Gly Gly Gly Ala Asp Gly Leu Thr
1               5

<210> SEQ ID NO 119
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-1

<400> SEQUENCE: 119

Arg Met Ile Val Asp Gly Leu Thr
1               5

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<210> SEQ ID NO 120
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-2

<400> SEQUENCE: 120

Lys Asn Gly Leu Asp Gly Leu Thr
1               5

<210> SEQ ID NO 121
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-3

<400> SEQUENCE: 121

Ala Leu Trp Ala Asp Gly Leu Thr
1               5

<210> SEQ ID NO 122
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-4

<400> SEQUENCE: 122

Gly Gly Gly Ala Ala Val Asp Gly
1               5

<210> SEQ ID NO 123
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-5

<400> SEQUENCE: 123

Gly Gly Gly Ala Met Pro Pro Gly
1               5

<210> SEQ ID NO 124
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-6

<400> SEQUENCE: 124

Gly Gly Gly Ala His Val Lys Trp
1               5

<210> SEQ ID NO 125
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-7

<400> SEQUENCE: 125

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Gly Gly Gly Ala Asp Ile Trp Ser
1 5

<210> SEQ ID NO 126
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-8
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 126

Tyr Gly Xaa Leu Asp Gly Leu Thr
1 5

<210> SEQ ID NO 127
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-9
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (1)..(1)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 127

Xaa Pro Ala Glu Asp Gly Leu Thr
1 5

<210> SEQ ID NO 128
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-10

<400> SEQUENCE: 128

Gly Val Gly Thr Asp Gly Leu Thr
1 5

<210> SEQ ID NO 129
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-11

<400> SEQUENCE: 129

Gly Gly Ala His Pro Ser Leu Thr
1 5

<210> SEQ ID NO 130
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-12

<400> SEQUENCE: 130

Gly Gly Ala Ala Val Trp Leu Thr
1 5

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<210> SEQ ID NO 131
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-13

<400> SEQUENCE: 131

Gly Gly Tyr Val Lys Leu Leu Thr
1               5


<210> SEQ ID NO 132
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-14
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 132

Gly Gly Xaa Leu Arg Trp Leu Thr
1               5


<210> SEQ ID NO 133
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-15

<400> SEQUENCE: 133

Gly Gly Gly Ala Glu Ile Val Ala
1               5


<210> SEQ ID NO 134
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-16

<400> SEQUENCE: 134

Gln Lys Ala Glu Asp Gly Leu Thr
1               5


<210> SEQ ID NO 135
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-17

<400> SEQUENCE: 135

Ala Leu Gly Ile Asp Gly Leu Thr
1               5


<210> SEQ ID NO 136
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-22
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<400> SEQUENCE: 136

Gly Val Gly Ser Asp Gly Leu Thr
1               5

<210> SEQ ID NO 137
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-24

<400> SEQUENCE: 137

Gly Val Gly Glu Asp Gly Leu Thr
1               5

<210> SEQ ID NO 138
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-26
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 138

Gly Gly Asp Ile Thr Xaa Leu Thr
1               5

<210> SEQ ID NO 139
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-27
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (8)..(8)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 139

Gly Gly Gly Ala Gly Trp Met Xaa
1               5

<210> SEQ ID NO 140
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR2-3alpha-28

<400> SEQUENCE: 140

Gly Gly Gly Ala Asp His Phe Cys
1               5

<210> SEQ ID NO 141
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-1

<400> SEQUENCE: 141

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Thr Leu Gln Phe Leu Tyr Asn Glu Gln Phe
1 5 10

<210> SEQ ID NO 142
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-2

<400> SEQUENCE: 142

Glu Gly Lys Trp Leu Tyr Asn Glu Gln Phe
1 5 10

<210> SEQ ID NO 143
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-3

<400> SEQUENCE: 143

Pro His Phe Leu Leu Tyr Asn Glu Gln Phe
1 5 10

<210> SEQ ID NO 144
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-4

<400> SEQUENCE: 144

Arg Thr Val Glu Leu Tyr Asn Glu Gln Phe
1 5 10

<210> SEQ ID NO 145
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-5

<400> SEQUENCE: 145

Ala Ser Val Gln Thr Thr Asn Glu Gln Phe
1 5 10

<210> SEQ ID NO 146
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-6
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 146

Ala Ser Pro Tyr Asp Xaa Asn Glu Gln Phe
1 5 10

<210> SEQ ID NO 147
<211> LENGTH: 10
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-7
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 147

Ala Ser Arg His Ser Arg Xaa Cys Gln Phe
1 5 10

<210> SEQ ID NO 148
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-8

<400> SEQUENCE: 148

Ala Ser Arg His Leu Tyr Phe Asp Asn Gly
1 5 10

<210> SEQ ID NO 149
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-9

<400> SEQUENCE: 149

Cys Ser Asn Thr Leu Tyr Asn Glu Gln Phe
1 5 10

<210> SEQ ID NO 150
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-10
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (2)..(3)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 150

Arg Xaa Xaa Pro Leu Tyr Asn Glu Gln Phe
1 5 10

<210> SEQ ID NO 151
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-11

<400> SEQUENCE: 151

Glu Arg Arg Pro Leu Tyr Asn Glu Gln Phe
1 5 10

<210> SEQ ID NO 152
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-12

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<400> SEQUENCE: 152

Ala Ser Trp Asn Val Ser Asn Glu Gln Phe
1               5                   10

<210> SEQ ID NO 153
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-13
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (3)..(3)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 153

Ala Ser Xaa Leu Leu Cys Asn Glu Gln Phe
1               5                   10

<210> SEQ ID NO 154
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-14

<400> SEQUENCE: 154

Ala Ser Tyr Val Val Ser Asn Glu Gln Phe
1               5                   10

<210> SEQ ID NO 155
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-15

<400> SEQUENCE: 155

Ala Ser His Pro Ala Arg Asn Glu Gln Phe
1               5                   10

<210> SEQ ID NO 156
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-16

<400> SEQUENCE: 156

Ala Ser Ala Asn Ala Val Asn Glu Gln Phe
1               5                   10

<210> SEQ ID NO 157
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-17

<400> SEQUENCE: 157

Ala Ser Gly Arg Val Arg Asn Glu Gln Phe
1               5                   10

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<210> SEQ ID NO 158
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-18

<400> SEQUENCE: 158

Ala Ser Arg His Leu Tyr Ser Ser Gly Arg
1               5                   10

<210> SEQ ID NO 159
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-19

<400> SEQUENCE: 159

Ala Ser Arg His Leu Tyr Pro Leu Tyr Ala
1               5                   10

<210> SEQ ID NO 160
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-26

<400> SEQUENCE: 160

Ala Ser Arg Arg Ser Phe Asn Glu Gln Phe
1               5                   10

<210> SEQ ID NO 161
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR3-3beta-31

<400> SEQUENCE: 161

Ala Ser Arg His Leu Tyr Pro Gly Ala Leu
1               5                   10

<210> SEQ ID NO 162
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-1

<400> SEQUENCE: 162

Asn Leu Asp Asp Gly Ile Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 163
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-2

<400> SEQUENCE: 163

Ser Cys Trp Ile Gly Ile Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

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<210> SEQ ID NO 164
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-3
<220> FEATURE:
<221> NAME/KEY: MOD_RES
<222> LOCATION: (1)..(1)
<223> OTHER INFORMATION: X is null by being coded as Amber Stop Condon

<400> SEQUENCE: 164

Xaa Pro Phe Leu Gly Ile Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15


<210> SEQ ID NO 165
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-4

<400> SEQUENCE: 165

Ala Ser Gly His Arg Ile Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15


<210> SEQ ID NO 166
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-5

<400> SEQUENCE: 166

Ala Ser Ser Gln Cys Gly Val Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15


<210> SEQ ID NO 167
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-6

<400> SEQUENCE: 167

Ala Ser Ser Gln Gly Ile Gly Leu Pro Glu Gly Arg Tyr Glu Gln Tyr
1               5                   10                  15


<210> SEQ ID NO 168
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-7

<400> SEQUENCE: 168

Ala Ser Ser Gln Gly Ile Gly Leu Ala Gly Gly Ile Glu Arg Ser Leu
1               5                   10                  15


<210> SEQ ID NO 169
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: TCR4-3beta-8

<400> SEQUENCE: 169

Ala Ser Ser Gln Gly Ile Gly Leu Ala Gly Gly Ile Arg Ser Lys Ile
1 5 10 15

<210> SEQ ID NO 170
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-9

<400> SEQUENCE: 170

Ala Ser Ala Leu Thr Arg Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 171
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-10

<400> SEQUENCE: 171

Ala Ser Ala Leu Ser Tyr Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 172
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-11

<400> SEQUENCE: 172

Ala Ser Ala Ile Thr Asp Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 173
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-12

<400> SEQUENCE: 173

Ala Ser Ser Val Thr Tyr Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 174
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-13

<400> SEQUENCE: 174

Ala Ser Ser Ile Thr Ser Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 175
<211> LENGTH: 16
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-14

<400> SEQUENCE: 175

Ala Ser Ser Val Thr Leu Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 176
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-15

<400> SEQUENCE: 176

Ala Ser Ser Leu Thr Tyr Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 177
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-16

<400> SEQUENCE: 177

Ala Ser Ser Met Thr Arg Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 178
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-17

<400> SEQUENCE: 178

Ala Ser Ser Val Thr Met Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 179
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-18

<400> SEQUENCE: 179

Ala Ser Ser Ile Thr Arg Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 180
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-19

<400> SEQUENCE: 180

Ala Ser Ser Gln Ser Arg Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1 5 10 15

<210> SEQ ID NO 181

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<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-20

<400> SEQUENCE: 181

Ala Ser Ser Gln Thr Ser Lys Cys Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 182
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-21

<400> SEQUENCE: 182

Ala Ser Ala Ile Thr Arg Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 183
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-22

<400> SEQUENCE: 183

Ala Ser Ala Val Thr Arg Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 184
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-23

<400> SEQUENCE: 184

Ala Ser Ala Leu Thr Ser Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 185
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-28

<400> SEQUENCE: 185

Ala Ser Ser Val Thr Arg Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

<210> SEQ ID NO 186
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-31

<400> SEQUENCE: 186

Ala Ser Ser Leu Thr His Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15

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<210> SEQ ID NO 187
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TCR4-3beta-32

<400> SEQUENCE: 187

Ala Ser Ser Leu Thr Thr Gly Leu Ala Gly Gly Ile Tyr Glu Gln Tyr
1               5                   10                  15


<210> SEQ ID NO 188
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A0B0 CDR3alpha

<400> SEQUENCE: 188

Ala Val Glu Gly Asp Ser Asn Tyr Gln Leu Ile
1               5                   10


<210> SEQ ID NO 189
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: A0B0 CDR3beta

<400> SEQUENCE: 189

Ala Ser Ser Leu Tyr Gly Gly Gly Asp Thr
1               5                   10
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**1**. A method for screening an αβ-TCR or a mutant thereof, comprising the construction of a phagemid, wherein, in the construction: a gene encoding A1 polypeptide and a gene encoding B1 polypeptide are inserted at the 3' ends of genes encoding constant regions of the α chain and the β chain of a TCR, respectively, and a ribosomal binding site and a molecular chaperone gene are integrated into a phage vector;

the amino acid sequence of the A1 polypeptide is as shown in SEQ ID NO: 1 or a variant thereof, the amino acid sequence of the B1 polypeptide is as shown in SEQ ID NO: 2 or a variant thereof; the variant retains at least the function of the unmutated polypeptide.

**2**. The method according to claim **1**, wherein, the constant region of the TCR α chain and the gene encoding the A1 polypeptide are linked by Linker 1; the amino acid sequence of Linker 1 is preferably as shown in SEQ ID NO: 3;

and/or, the constant region of the TCR β chain and the gene encoding the B1 polypeptide are linked by Linker 2; the amino acid sequence of Linker 2 is preferably as shown in SEQ ID NO: 3;

and/or, the amino acid sequence of the constant region of the TCR α chain is as shown in SEQ ID NO: 5;

and/or, the amino acid sequence of the constant region of the TCR β chain is as shown as SEQ ID NO: 4 or SEQ ID NO: 6, or a variant thereof, the variant of the sequence as shown in SEQ ID NO: 6 preferably comprises mutation sites C75A and N89D.

**3**. The method according to claim **1**, wherein, the ribosome binding site and the molecular chaperone gene are located at the 3' end of the gene expressing a capsid protein, the capsid protein is preferably a pIII protein;

and/or, the nucleotide sequence of the ribosomal binding site is as shown in SEQ ID NO: 7;

and/or, the molecular chaperone gene is skp gene, the nucleotide sequence of the skp gene is preferably as shown in SEQ ID NO: 8.

**4**. The method according to claim **1**, wherein, the backbone of the phage vector is pCANTAB5E;

preferably, in the pCANTAB5E, the pelB signal peptide gene and/or the Myc tag gene are introduced to replace the sequence between the gIII signal and the Amber Stop Codon;

more preferably, the nucleotide sequence between the pelB signal peptide gene and the Amber Stop Codon (both are comprised) is as shown in SEQ ID NO: 9.

**5**. The method according to claim **1**, wherein, the gene of the TCR α chain and the gene of the TCR β chain are located in two different expression frames;

preferably, the phagemid comprises an expression element expressing following polypeptide (1) and polypeptide (2),

polypeptide (1): pelB signal peptide-variable region of TCR α chain (Vα)-constant region of α chain (Cα)-Linker 1-A1 peptide, polypeptide (2): gIII signal peptide-variable region of TCR β chain (Vβ)-constant region of β chain (Cβ)-Linker 2-B1 peptide-pIII protein;

more preferably, the cysteine residue located behind the constant region of the TCR α chain and the constant region of the TCR β chain, and the cysteine located before the N-terminal portion of the transmembrane region is subject to substitution or deletion.

**6**. A TCR obtained by screening with the method according to claim **1**, wherein, the α chain of the TCR comprises:

(1) CDRs 1-3 or a variant thereof of the α chain variant region having the amino acid sequence as shown in SEQ ID NO: 10, wherein: the amino acid sequence of the CDR3 variant is preferably as shown in SEQ ID NO: 19;

(2) CDRs 1-3 or a variant thereof of the α chain variant region having the amino acid sequence as shown in SEQ ID NO: 14; or

(3) CDRs 1-3 or a variant thereof of the α chain variant region having the amino acid sequence as shown in SEQ ID NO: 16.

**7**. The TCR according to claim **6**, wherein, the β chain of the TCR comprises:

(1) CDRs 1-3 or a variant thereof of the β chain variable region having the amino acid sequence as shown in SEQ ID NO: 11, wherein: the variant is obtained by position 6 and/or position 8 of CDR3 being substituted, position 6 is preferably substituted with Y or F, position 8 is preferably substituted with R, T or Q; preferably, the sequence of the variant is as shown in any one of SEQ ID NOs: 21-24;

(2) CDRs 1-3 or a variant thereof of the β chain variable region having the amino acid sequence as shown in SEQ ID NO: 15, wherein, the sequence of the variant is preferably as shown in any one of SEQ ID NOs: 26-30;

or (3) CDRs 1-3 or a variant thereof of the β chain variable region having the amino acid sequence as shown in SEQ ID NO: 17, wherein, a mutation occurs in 1 site, 2 sites, 3 sites or 4 sites of positions 2-6 of CDR3; when the mutation occurs in two or more sites, the two or more sites are contiguous; position 3 is preferably mutated to A, position 4 is preferably mutated to Q, position 5 is preferably mutated to G, position 6 is preferably mutated to S, R or W; preferably, the sequence of the variant is as shown in any one of SEQ ID NOs: 32-39.

**8**. A phage vector for screening an αβ-TCR heterodimer or a mutant thereof, wherein, the phage vector comprises an expression element expressing following polypeptide (1) and polypeptide (2):

polypeptide (1): pelB signal peptide-variable region of the α chain-constant region of the α chain-Linker 1-A1 peptide;

polypeptide (2): gIII signal peptide-variable region of the β chain-constant region of the β chain-Linker 2-B1 peptide-pIII protein;

preferably:

the amino acid sequence of the constant region of the α chain ($C_\alpha$) is as shown in SEQ ID NO: 5;

and/or, the amino acid sequence of the constant region of the β chain ($C_\beta$) is as shown in SEQ ID NO: 4 or SEQ ID NO: 6;

and/or, the amino acid sequence of Linker 1 is as shown in SEQ ID NO: 3;

and/or, the amino acid sequence of Linker 2 is as shown in SEQ ID NO: 3;

and/or, the amino acid sequence of A1 peptide is as shown in SEQ ID NO: 1;

and/or, the amino acid sequence of B1 peptide is as shown in SEQ ID NO: 2;

and/or, the backbone of the phage vector is pCANTABSE;

and/or, a ribosomal binding site and a molecular chaperone gene are integrated into the phage vector, such as, integrated at the 3' end of the gene expressing a capsid protein; the molecular chaperone gene is preferably the skp gene, more preferably is the sequence as shown in SEQ ID NO: 8; the sequence of the ribosomal binding site is preferably as shown in SEQ ID NO: 7;

and/or, the amino acid sequence of the constant region of the α chain is as shown in SEQ ID NO: 5;

and/or, the amino acid sequence of the constant region of the β chain is as shown in SEQ ID NO: 4 or SEQ ID NO: 6, or a variant thereof, the variant of the sequence as shown in SEQ ID NO: 6 preferably comprises mutation sites C75A and N89D.

**9**. A method for screening a TCR mutant by using the polypeptide as shown in SEQ ID NO: 1 or 2.

**10**. The method according to claim **9**, wherein, during the use, the gene encoding the polypeptide is placed at the 3' end of the gene of constant region of the α chain or the β chain of the TCR, respectively;

preferably, the gene encoding the polypeptide is linked to the gene of the constant region of the α chain or the β chain of the TCR by a linker; the amino acid sequence of the linker is preferably as shown in SEQ ID NO: 3.

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